Pacific Herring preliminary data summary for West Coast of Vancouver Island 2026

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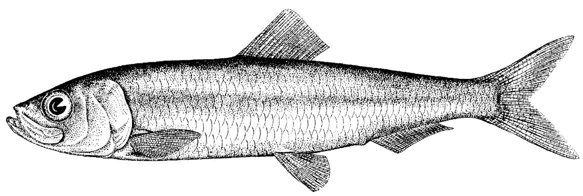

Pacific Herring (*Clupea pallasii*). Image credit: Fisheries and Oceans Canada.

Disclaimer This report contains preliminary data collected for Pacific Herring in 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). These data may differ from data used and presented in the final stock assessment.

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1 Context

Pacific Herring (*Clupea pallasii*) in British Columbia are assessed as 5 major and 2 minor stock assessment regions (SARs), and data are collected and summarized on this scale (Table 1, Figure 1). Note that formal stock assessments are only done for major SARs. The Pacific Herring data collection program includes fishery-dependent and -independent data from 1951 to 2026. This includes annual time series of commercial catch data, biological samples (providing information on proportion-at-age and weight-at-age), and spawn index data conducted using a combination of surface and SCUBA surveys. In some areas, industry- and/or First Nations-operated in-season soundings programs are also conducted, and this information is used by resource managers, First Nations, and

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stakeholders to locate fish and identify areas of high and low Pacific Herring biomass to plan harvesting activities. In-season acoustic soundings are not used by stock assessment to inform the estimation of spawning biomass.

The following is a description of data collected for Pacific Herring in 2026 in the West Coast of Vancouver Island major SAR (Figure 2). Data collected outside the SAR boundary are not included in this summary, and are not used for the purposes of stock assessment. Although we summarise data at the scale of the SAR for stock assessments, we summarise data at finer spatial scales in this report: Locations are nested within Sections, Sections are nested within Statistical Areas, and Statistical Areas are nested within SARs (Table 2). Note that we refer to 'year' instead of 'herring season' in this report; therefore 2026 refers to the 2025/2026 Pacific Herring season.

2 Data collection programs

Biological samples were collected by the seine charter vessel *Ocean Achiever* for 15 days from February 20th to March 6th. The primary purpose of the test charter vessel was to collect biological samples from main bodies of herring from Statistical Areas 23, 24, and 25. Nearshore herring samples were collected by the Nuu-chah-nulth staff as part of a sampling program and on-going collaboration between WCVI First Nations and DFO. These nearshore biological samples were collected from spawning aggregations using cast nets. WCVI First Nations spawn reconnaissance charters reported spawn activity in all three areas. Herring spawn locations were primarily identified with fixed-wing overflights conducted by DFO Resource Management area staff. Ten flights were conducted this season all in March.

One dive charter vessel and a shorebased crew operated in WCVI:

- The *Pachena No. 1* surveyed 18 days from March 5th to March 22nd, and
- The shorebased dive surveyed 17 days from February 27th to March 15th.

3 Catch and biological samples

In the 1950s and 1960s, the reduction fishery dominated Pacific Herring catch; starting in the 1970s, catch has been predominantly from roe seine and gillnet fisheries. The reduction fishery is different from current fisheries in several ways. First, the reduction fishery caught Pacific Herring of all ages, whereas current fisheries target spawning (i.e., mature) fish. Thus, reduction fisheries included an unknown number of age-1 fish which are not typically caught in current fisheries. Second, the reduction fishery has some uncertainty regarding the quantity and location of catch; however catches have been resolved to SAR and Statistical Area using fish slips as best as possible. For the roe gillnet fishery, all Pacific Herring catch has been validated by a dockside monitoring program since 1998; the catch validation program started in 1999 for the roe seine fishery. Finally, the reduction fishery operated during the summer and winter months, whereas roe fisheries typically target spawning fish between February and April.

Landed commercial catch of Pacific Herring by year and fishery is shown in Table 3 and Figure 3. Total harvested spawn-on-kelp (SOK) in 2026 in the West Coast of Vancouver Island major SAR is shown in Table 4; we also calculate the estimated spawning biomass associated with SOK harvest. See the spawn index technical report for calculations to convert SOK harvest to spawning biomass. Incidental mortality from fisheries and aquaculture activities is shown in Figure 4.

In 2026, 35 Pacific Herring biological samples were collected and processed for the West Coast of Vancouver Island major SAR (Table 5, Table 6). The nearshore sampling program is a multi-year pilot study using cast nets (Figure 10, Figure 11, and Figure 12). Differences between biological data collected from two sampling protocols regarding the number-, proportion-, weight-, and length-at-age for Pacific Herring in 2026 in the West Coast of Vancouver Island major SAR are shown in Table 7, Table 8, Table 9, and Table 10, respectively. Summaries of data collected by the nearshore sampling program are shown in Table 11, Table 12, Table 13, and Table 14. We compare length-, weight, and proportion-at-age as well as length-weight relationships for nearshore and seine test samples in Figure 13, Figure 14, Figure 15, and Figure 16, respectively. Note that only biological data from the seine samples are used for the purposes of stock assessment. The locations in which the biological samples were collected are presented in Figure 5. Included herein are biological summaries of observed proportion-, number-, weight-, and length-at-age (Figure 6, Table 15, and Figure 7, respectively). We also show the percent change in weight and length for age-3 and age-6 fish (Figure 8 & Figure 9, respectively). Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. Only representative biological samples are included, where 'representative' indicates whether the Pacific Herring sample in the set accurately reflects the larger Pacific Herring school.

3.1 Nearshore genetics project

DFO initiated a nearshore genetics project in 2022 to better understand the spatial and temporal structure of Pacific Herring stocks in British Columbia. We do this by collecting genetic and biological data from actively spawning Pacific Herring (*Clupea pallasii*). Then we collect biological data (e.g., length, weight, age, sex) and genetic data from the fish. Finally, we analyse the biological and genetic data to better understand Pacific Herring. Genetic samples were collected in the West Coast of Vancouver Island major SAR in 2024 and 2025 (Figure 17). Results from the nearshore genetics project will be presented in a separate document when they are available.

4 Spawn survey data

Pacific Herring spawn surveys were conducted at 40 individual locations in 2026 in the West Coast of Vancouver Island major SAR (Table 16, and Figure 18). A summary of spawn from the last decade (2016 to 2025) is shown in Figure 19. Figure 20 shows spawn start date by decade and Statistical Area. Spawn surveys are conducted to estimate the spawn length, width, number of egg layers, and substrate type, and these data are used to estimate the index of spawning biomass (i.e., the spawn index; Figure 21, Figure 22, Figure 23, Figure 24, Table 17, and Figure 25). See the spawn index technical report for calculations to convert SOK harvest to spawning biomass. The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Therefore, these data do not represent model estimates of spawning biomass, and are considered the minimum observed spawning biomass derived from egg counts. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

Some Pacific Herring Sections contribute more than others to the total spawn index, and the percentage contributed by Section varies yearly (Figure 24b, Figure 26). For example, in 2026, Section 245 contributed the most to the spawn index (48%). As with Sections, some Statistical Areas contribute more than others to the total spawn index (Figure 24c, Figure 27). An animation shows the spawn index by spawn survey location from 1951 to 2026 (Figure 28).

5 First Nations observations

The following observations were contributed by representatives of First Nations communities. These observations provide context and perspectives to this data report. In some cases we make edits for clarity and brevity, but we do not change the intent or substance of responses. Note: if your observations are missing, submit them here.

5.1 Nuu-chah-nulth Tribal Council

Managed herring spawn reconnaissance monitoring contract for PFMAs 24 and 25 and assisted Nations in Area 23 including collecting biosamples. Biggest challenge is having monitors on the grounds at the right times and communications between monitors.

A few good spawns, most short in duration. Increased spawn distribution compared to 2025, but still truncated compared to the 1990's. Some good SOB harvests. Broad distribution of spawn in Area 24, several multiple days of spawn. Numerous good SOB harvests in Area 24. Area 23: spawn duration short, and some spawn in new areas such as near Bamfield and Alma Russel Islands. SOB harvest spotty in Area 23.

Caught whole herring from jigging and some off the test boat. Short durations of spawn in Area 23 and 25 make getting SOB more challenging as the time to find the spawn and set boughs is short.

6 General observations

The following observations were reported by DFO Resource Management staff and DFO Science staff. These observations provide additional context to this data report.

- Active spawn reported, observed, or identified via satellite image between February 16th in Hesquiat Harbour and April 11th in Nootka Sound.
- Vessel- and shore-based dive surveys were performed on all observed spawns except for Clifford Point, Wickaninnish Island, early/late Hesquiat Harbour, outside of Nootka Island, late Nootka Sound, and Area 27.
- Spawn timing was widespread.

7 Tables

Table 1. Pacific Herring stock assessment regions (SARs) in British Columbia.

Name	Code	Type
Haida Gwaii	HG	Major
Prince Rupert District	PRD	Major
Central Coast	CC	Major
Strait of Georgia	SoG	Major
West Coast of Vancouver Island	WCVI	Major
Area 27	A27	Minor
Area 2 West	A2W	Minor

Table 2. Statistical Areas and Sections for Pacific Herring in the West Coast of Vancouver Island major stock assessment region (SAR).

Region	Statistical Area	Section
West Coast of Vancouver Island	23	230
West Coast of Vancouver Island	23	231
West Coast of Vancouver Island	23	232
West Coast of Vancouver Island	23	233
West Coast of Vancouver Island	23	239
West Coast of Vancouver Island	24	240
West Coast of Vancouver Island	24	241
West Coast of Vancouver Island	24	242
West Coast of Vancouver Island	24	243
West Coast of Vancouver Island	24	244
West Coast of Vancouver Island	24	245
West Coast of Vancouver Island	24	249
West Coast of Vancouver Island	25	250
West Coast of Vancouver Island	25	251
West Coast of Vancouver Island	25	252
West Coast of Vancouver Island	25	253
West Coast of Vancouver Island	25	259

Table 3. Total landed commercial catch of Pacific Herring in metric tonnes (t) by gear type in 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). Legend: 'Other' represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; 'RoeSN' represents the roe seine fishery; and 'RoeGN' represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Note: data may be withheld due to privacy concerns (WP).

Gear	Catch (t)
Other	0
RoeSN	0
RoeGN	0

Table 4. Total harvested Pacific Herring spawn-on-kelp (SOK) in pounds (lb), and the associated estimate of spawning biomass in metric tonnes (t) from 2016 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). See the spawn index technical report for calculations to convert SOK harvest to spawning biomass. Note: data may be withheld due to privacy concerns (WP).

Year	Harvest (lb)	Spawning biomass (t)
2016	0	0
2017	0	0
2018	0	0
2019	0	0
2020	0	0
2021	0	0
2022	0	0
2023	0	0
2024	0	0
2025	30,235	45
2026	54,631	81

Table 5. Number of Pacific Herring biological samples processed from 2016 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). Each sample is approximately 100 fish. Note: Nearshore samples are not used in stock assessments.

	Number of samples			
Year	Commercial	Test	Nearshore	Total
2016	0	15	10	25
2017	0	12	7	19
2018	0	20	22	42
2019	0	14	11	25
2020	0	12	7	19
2021	0	13	9	22
2022	0	10	19	29
2023	0	9	21	30
2024	0	15	23	38
2025	0	16	18	34
2026	0	9	26	35

Table 6. Number and type of Pacific Herring biological samples processed in 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). Each sample is approximately 100 fish.

Type	Gear	Use	Number of samples
Test	Other	Nearshore	26
Test	Seine	Test fishery	9

Table 7. Observed number-at-age of Pacific Herring by sample type in 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a 'plus group' which includes fish ages 10 and older. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

	Number-at-age								
Sample type	2	3	4	5	6	7	8	9	10
Nearshore	158	641	402	311	405	144	73	11	3
Seine test	72	256	155	96	132	48	15	3	1
Total	230	897	557	407	537	192	88	14	4

Table 8. Observed proportion-at-age of Pacific Herring by sample type in 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a 'plus group' which includes fish ages 10 and older. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

	Proportion-at-age								
Sample type	2	3	4	5	6	7	8	9	10
Nearshore	0.074	0.298	0.187	0.145	0.189	0.067	0.034	0.005	0.001
Seine test	0.093	0.329	0.199	0.123	0.170	0.062	0.019	0.004	0.001
Total	0.079	0.307	0.190	0.139	0.184	0.066	0.030	0.005	0.001

Table 9. Observed mean weight-at-age in grams (g) of Pacific Herring by sample type in 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a 'plus group' which includes fish ages 10 and older. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

	Mean weight-at-age (g)								
Sample type	2	3	4	5	6	7	8	9	10
Nearshore	59	71	83	104	116	119	122	135	144
Seine test	61	76	87	111	125	126	140	156	139
Total	60	72	84	106	118	120	125	140	142

Table 10. Observed mean length-at-age in millimetres (mm) of Pacific Herring by sample type in 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a 'plus group' which includes fish ages 10 and older. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

	Mean length-at-age (mm)								
Sample type	2	3	4	5	6	7	8	9	10
Nearshore	157	166	176	189	196	199	200	205	209
Seine test	155	168	176	191	197	199	204	209	203
Total	156	167	176	189	196	199	201	206	207

Table 11. Observed number-at-age of Pacific Herring for nearshore samples from 2015 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

	Number-at-age								
Year	2	3	4	5	6	7	8	9	10
2015	67	64	130	10	3	3	0	0	0
2016	32	537	161	122	13	3	2	0	0
2017	43	56	355	89	64	4	0	0	0
2018	171	203	142	1157	196	105	18	0	1
2019	101	616	82	37	106	15	6	0	0
2020	168	161	179	28	8	34	6	2	2
2021	47	479	146	90	11	5	8	1	1
2022	138	465	656	160	136	16	7	11	4
2023	23	798	484	384	105	52	8	1	3
2024	49	726	1348	493	313	90	10	3	3
2025	73	257	392	490	209	97	8	4	1
2026	158	641	402	311	405	144	73	11	3

Table 12. Observed proportion-at-age of Pacific Herring for nearshore samples from 2015 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

	Proportion-at-age								
Year	2	3	4	5	6	7	8	9	10
2015	0.242	0.231	0.469	0.036	0.011	0.011	0.000	0.000	0.000
2016	0.037	0.617	0.185	0.140	0.015	0.003	0.002	0.000	0.000
2017	0.070	0.092	0.581	0.146	0.105	0.007	0.000	0.000	0.000
2018	0.086	0.102	0.071	0.581	0.098	0.053	0.009	0.000	0.001
2019	0.105	0.640	0.085	0.038	0.110	0.016	0.006	0.000	0.000
2020	0.286	0.274	0.304	0.048	0.014	0.058	0.010	0.003	0.003
2021	0.060	0.608	0.185	0.114	0.014	0.006	0.010	0.001	0.001
2022	0.087	0.292	0.412	0.100	0.085	0.010	0.004	0.007	0.003
2023	0.012	0.429	0.260	0.207	0.057	0.028	0.004	0.001	0.002
2024	0.016	0.239	0.444	0.162	0.103	0.030	0.003	0.001	0.001
2025	0.048	0.168	0.256	0.320	0.137	0.063	0.005	0.003	0.001
2026	0.074	0.298	0.187	0.145	0.189	0.067	0.034	0.005	0.001

Table 13. Observed mean weight-at-age in grams (g) of Pacific Herring for nearshore samples from 2015 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a 'plus group' which includes fish ages 10 and older. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

	Mean weight-at-age (g)								
Year	2	3	4	5	6	7	8	9	10
2015	51	67	76	81	109	104	NA	NA	NA
2016	49	66	72	81	88	83	88	NA	NA
2017	58	70	85	91	94	98	NA	NA	NA
2018	53	77	85	95	103	107	112	NA	114
2019	54	70	81	100	111	116	118	NA	NA
2020	56	69	85	94	114	125	110	131	120
2021	57	71	82	101	111	90	124	121	121
2022	57	71	89	100	118	126	123	128	148
2023	44	72	84	97	108	119	123	142	122
2024	50	68	85	97	103	115	104	114	125
2025	58	71	91	108	118	121	141	129	119
2026	59	71	83	104	116	119	122	135	144

Table 14. Observed mean length-at-age in millimetres (mm) of Pacific Herring for nearshore samples from 2015 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a 'plus group' which includes fish ages 10 and older. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

	Mean length-at-age (mm)								
Year	2	3	4	5	6	7	8	9	10
2015	154	170	178	182	199	200	NA	NA	NA
2016	154	171	178	186	188	197	192	NA	NA
2017	166	178	190	193	197	201	NA	NA	NA
2018	159	179	188	196	200	202	211	NA	220
2019	155	170	179	191	198	203	201	NA	NA
2020	156	169	180	186	200	205	197	214	204
2021	156	172	180	193	202	188	208	209	197
2022	165	176	189	196	205	209	208	217	222
2023	146	169	179	187	194	198	200	212	208
2024	154	170	183	191	196	201	200	203	219
2025	157	168	181	192	197	200	207	202	204
2026	157	166	176	189	196	199	200	205	209

Table 15. Observed proportion-at-age for Pacific Herring from 2016 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The age-10 class is a 'plus group' which includes fish ages 10 and older.

	Proportion-at-age								
Year	2	3	4	5	6	7	8	9	10
2016	0.040	0.648	0.168	0.124	0.014	0.003	0.004	0.001	0.000
2017	0.025	0.076	0.664	0.132	0.082	0.016	0.002	0.002	0.000
2018	0.053	0.133	0.109	0.566	0.096	0.035	0.005	0.002	0.000
2019	0.055	0.504	0.097	0.049	0.232	0.040	0.018	0.003	0.000
2020	0.180	0.275	0.347	0.047	0.024	0.100	0.022	0.004	0.000
2021	0.173	0.533	0.136	0.118	0.014	0.009	0.013	0.003	0.002
2022	0.073	0.292	0.436	0.089	0.097	0.005	0.001	0.005	0.001
2023	0.065	0.501	0.222	0.146	0.039	0.019	0.004	0.004	0.000
2024	0.006	0.221	0.487	0.160	0.100	0.022	0.003	0.001	0.001
2025	0.033	0.183	0.223	0.345	0.125	0.075	0.015	0.001	0.000
2026	0.093	0.329	0.199	0.123	0.170	0.062	0.019	0.004	0.001

Table 16. Pacific Herring spawn survey locations, start date, and spawn index in metric tonnes (t) in 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Missing spawn index values indicate incomplete surveys (NAs).

Statistical Area	Section	Location name	Start date	Spawn index (t)
23	231	Eagle Bay	February 19	28
23	232	Forbes Is	February 19	473
23	232	Macoah Pass	February 19	1,086
23	232	Newcombe Chnl	February 20	886
23	232	Spilling Islet	February 20	66
23	232	Two Rivers	February 19	3,615
23	233	Alma Russell Is (East)	March 01	1,135
23	233	Alma Russell Is (West)	March 01	350
23	233	Robinson Is	March 01	66
24	242	Antons Spit	February 16	NA
24	242	Hesquiat Hrbr Hd	February 24	195
24	242	Leclaire Pt	February 24	550
24	242	Rondeault Pt	February 16	87
24	244	Matilda Inlt	March 05	479
24	244	McNeil Pen	March 05	1,120
24	245	Burgess Islets	March 06	1,312
24	245	Calmus Pass	March 05	1,316
24	245	Chetarpe Reserve	March 02	746
24	245	Elbow Bank	March 03	8,961
24	245	Epper Pass	March 02	1,293
24	245	Father Charles Chnl	March 02	4
24	245	Kakawis	March 03	12
24	245	Kutcous Islets	March 02	16
24	245	Maurus Chnl	March 03	308
24	245	McIntosh Bay	March 06	1,059
24	245	Opitsat Reserve	March 02	99
24	245	Robert Pt	March 03	386
24	245	Stubbs Is	March 02	1,550
24	245	Whitesand Cv	March 02	1,587
24	245	Yarksis Reserve	March 02	2,893
25	252	Boca del Infierno Bay	April 11	NA
25	252	Friendly Cv	March 02	632
25	252	Santa Gertrudis Cv	March 02	53
25	253	Catala Is	February 27	3,525
25	253	Nuchatlitz Inner	February 25	3,235
25	253	Nuchatlitz Outer	February 25	2,167
25	253	Owossitsa Cr	February 27	158
25	253	Port Langford	February 24	1,942

Table 16 continued

Statistical Area	Section	Location name	Start date	Spawn index (t)
25	253	Rosa Hrbr	February 25	1,007
25	253	Rosa Is	February 25	889

Table 17. Summary of Pacific Herring spawn survey data from 2016 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Units: metres (m), and metric tonnes (t).

Year	Total length (m)	Mean width (m)	Mean number of egg layers	Spawn index (t)
2016	60,575	122	1.4	20,528
2017	44,200	180	1.3	16,476
2018	61,825	136	1.5	28,107
2019	49,325	130	1.2	17,029
2020	44,125	162	0.9	18,760
2021	45,851	203	1.1	29,338
2022	64,925	168	1.3	23,706
2023	74,850	195	1.9	77,005
2024	102,550	160	2.5	86,307
2025	117,950	134	1.6	47,525
2026	89,400	130	1.8	45,303

8 Figures

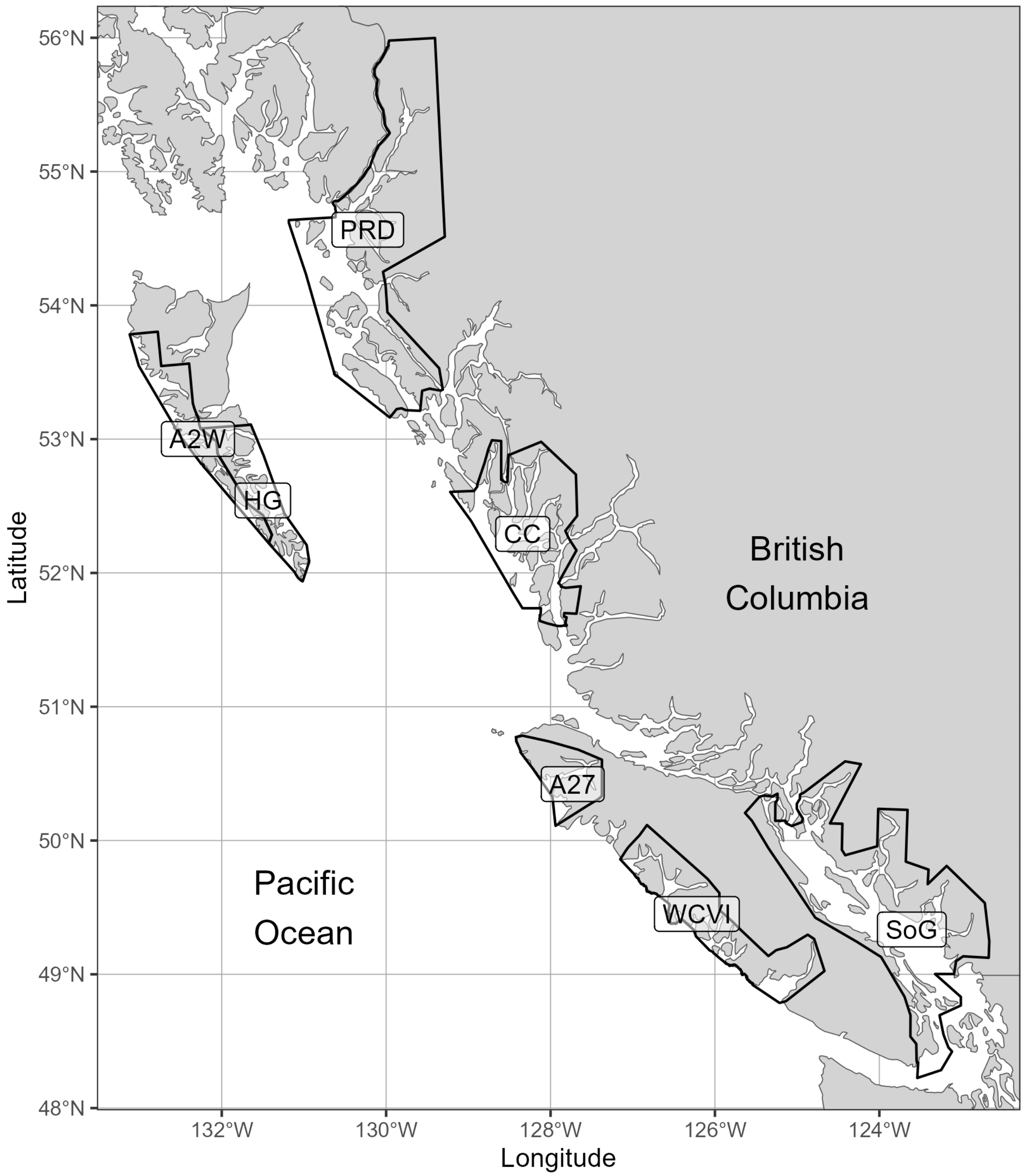

Figure 1. Boundaries for the Pacific Herring stock assessment regions (SARs) in British Columbia. There are 5 major SARs: Haida Gwaii (HG), Prince Rupert District (PRD), Central Coast (CC), Strait of Georgia (SoG), and West Coast of Vancouver Island (WCVI). There are 2 minor SARs: Area 27 (A27) and Area 2 West (A2W). Units: kilometres (km).

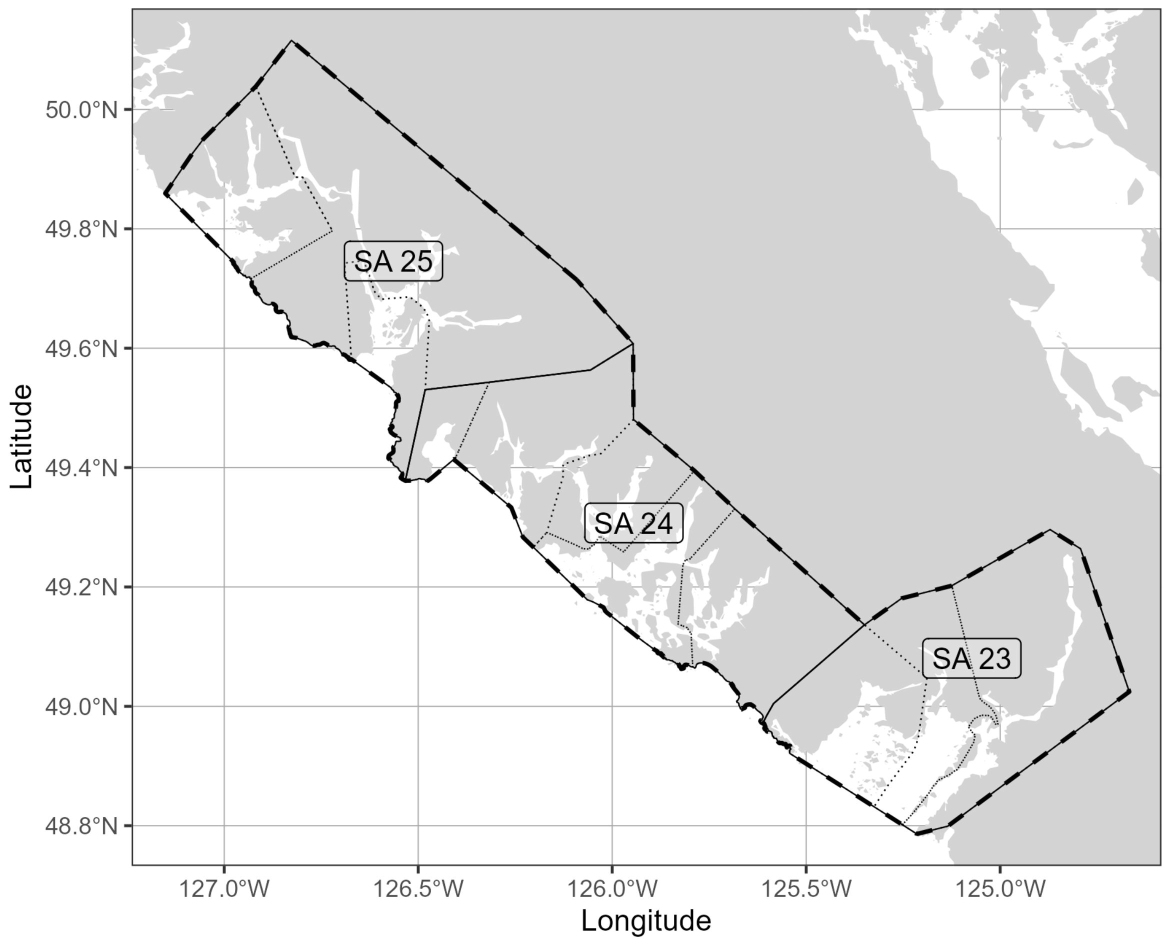

Figure 2. Boundaries for the West Coast of Vancouver Island major stock assessment region (SAR; thick dashed lines), associated Statistical Areas (SA; thin solid lines), and associated Sections (thin dotted lines). Units: kilometres (km).

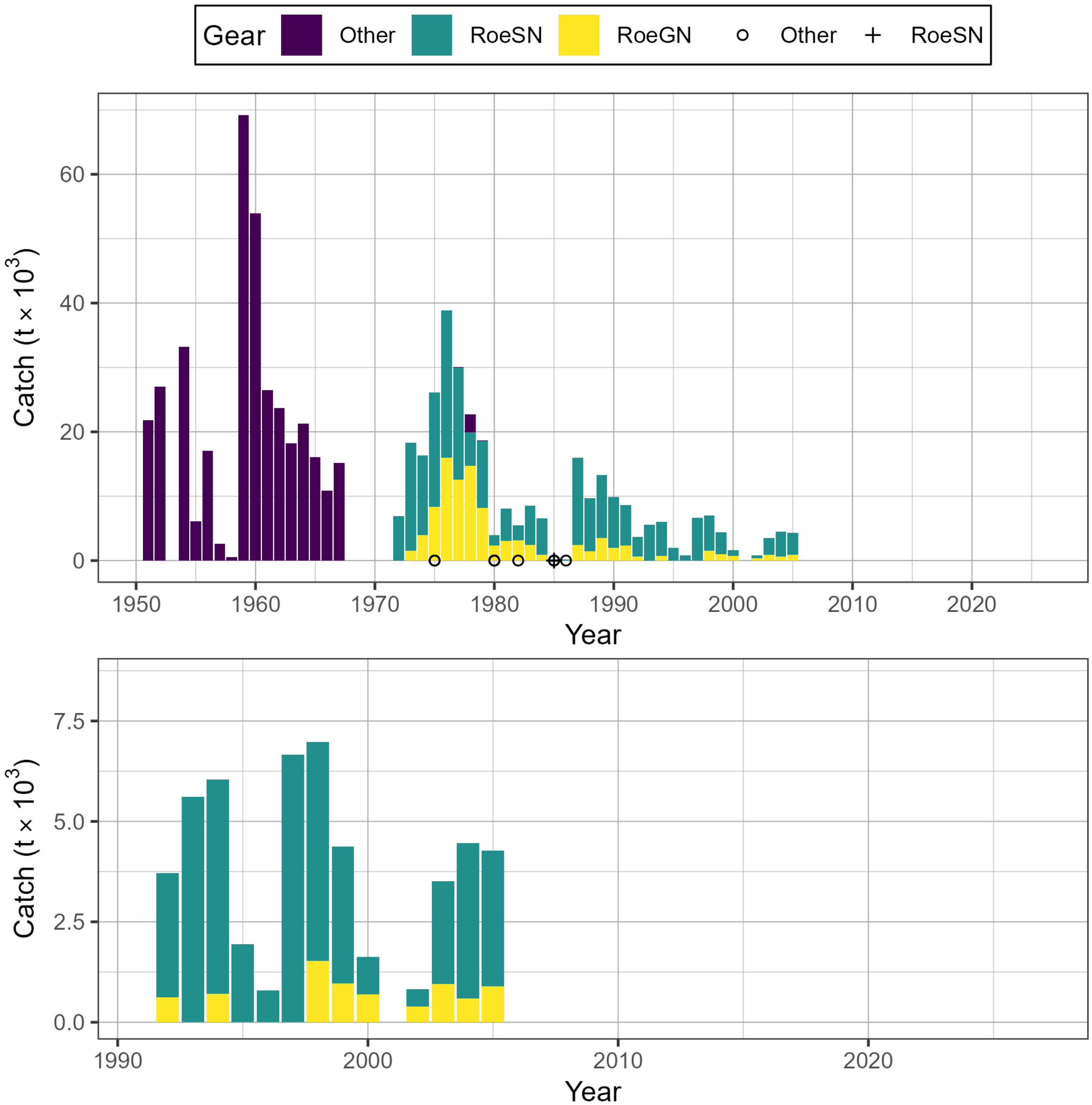

Figure 3. Time series of total landed catch in thousands of metric tonnes (t $\times$ 10^3) of Pacific Herring by gear type from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). Legend: 'Other' represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; 'RoeSN' represents the roe seine fishery; and 'RoeGN' represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Bottom panel shows catch since 1991 in more detail. Note: symbols indicate years in which catch by gear type (i.e., Other, RoeSN, RoeGN) is withheld due to privacy concerns.

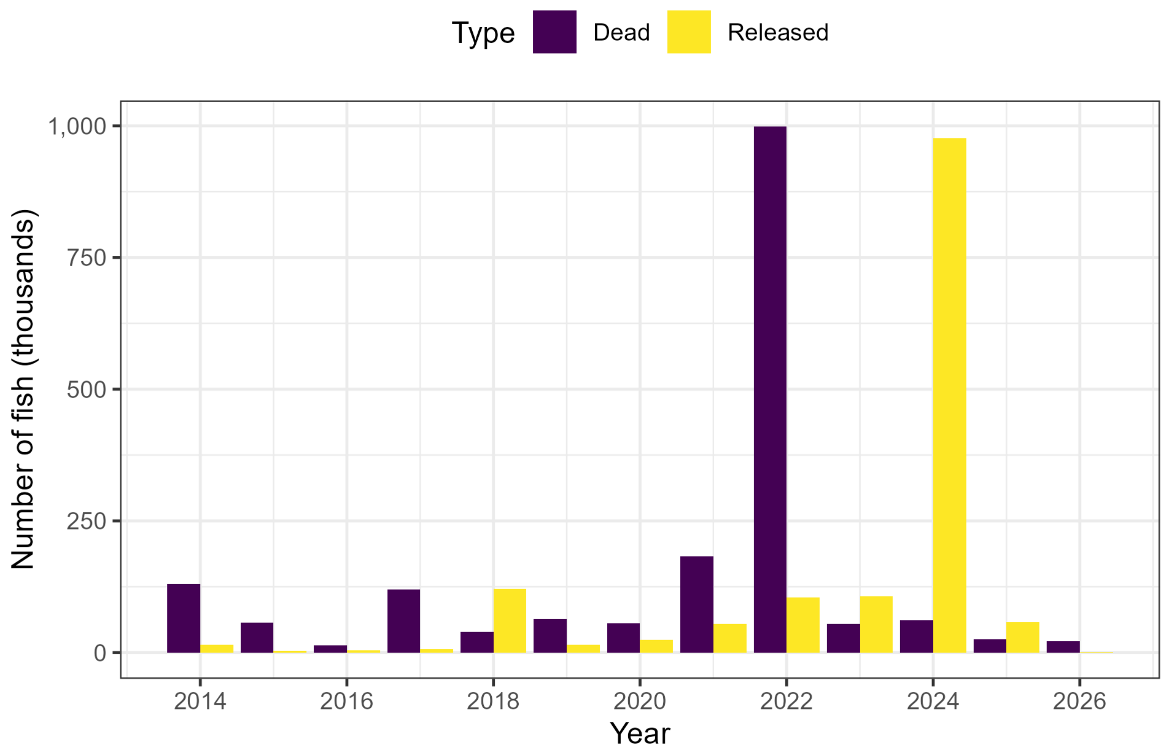

Figure 4. Incidental Pacific Herring mortality in aquaculture activities in thousands of fish from 2014 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). Note: figure may include data outside SAR boundaries.

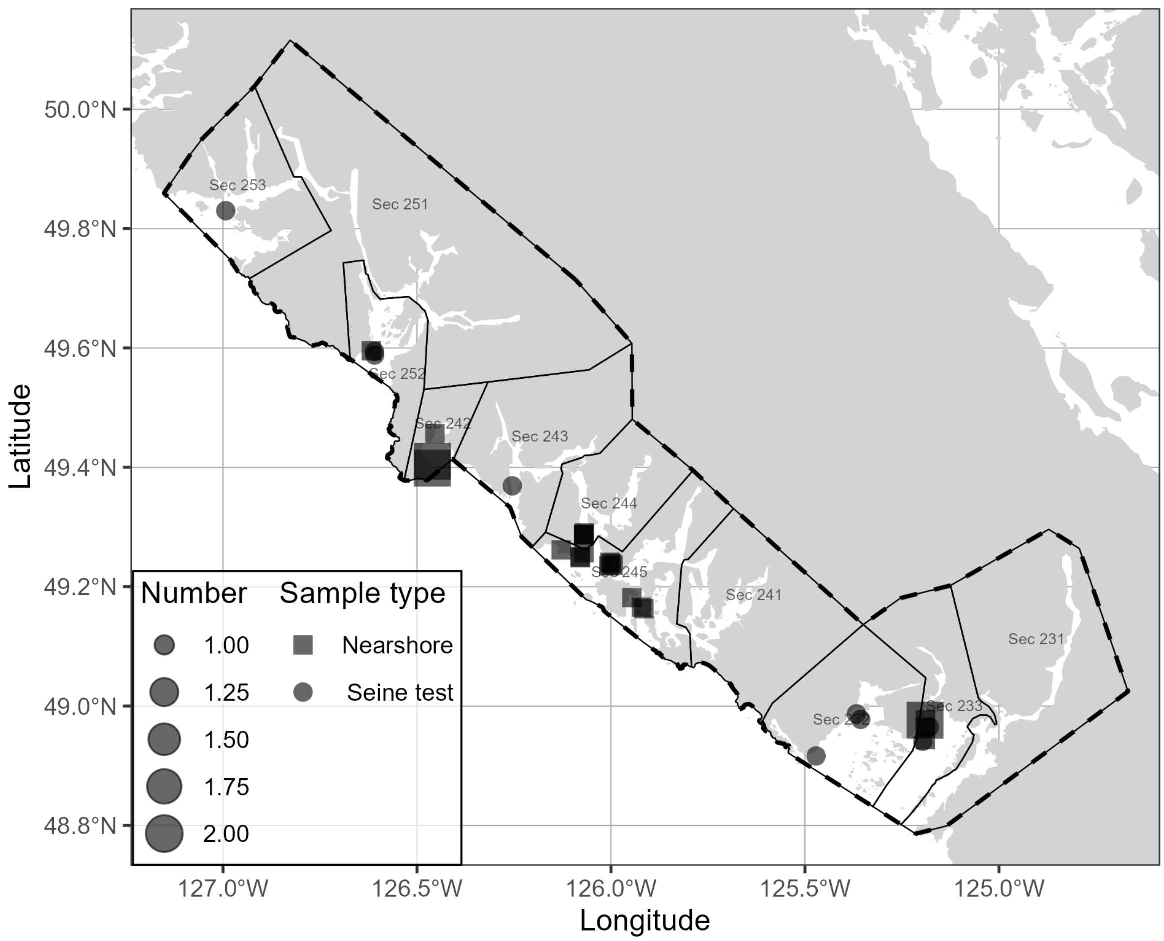

Figure 5. Location and type of Pacific Herring biological samples collected in 2026 in the West Coast of Vancouver Island major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). Units: kilometres (km).

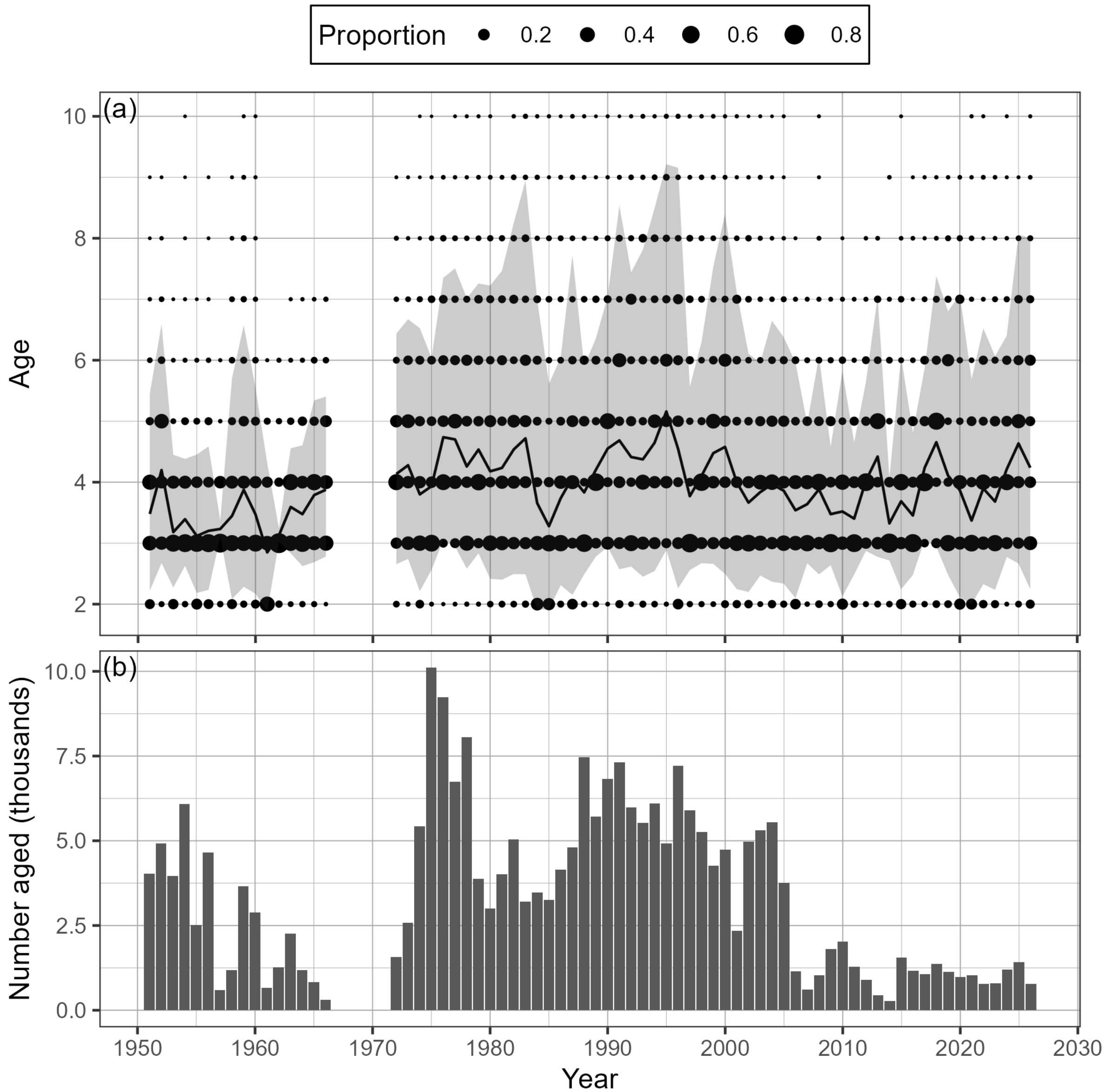

Figure 6. Time series of observed proportion-at-age (a) and number aged in thousands (b) of Pacific Herring from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The black line is the mean age, and the shaded area is the approximate 90% distribution. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

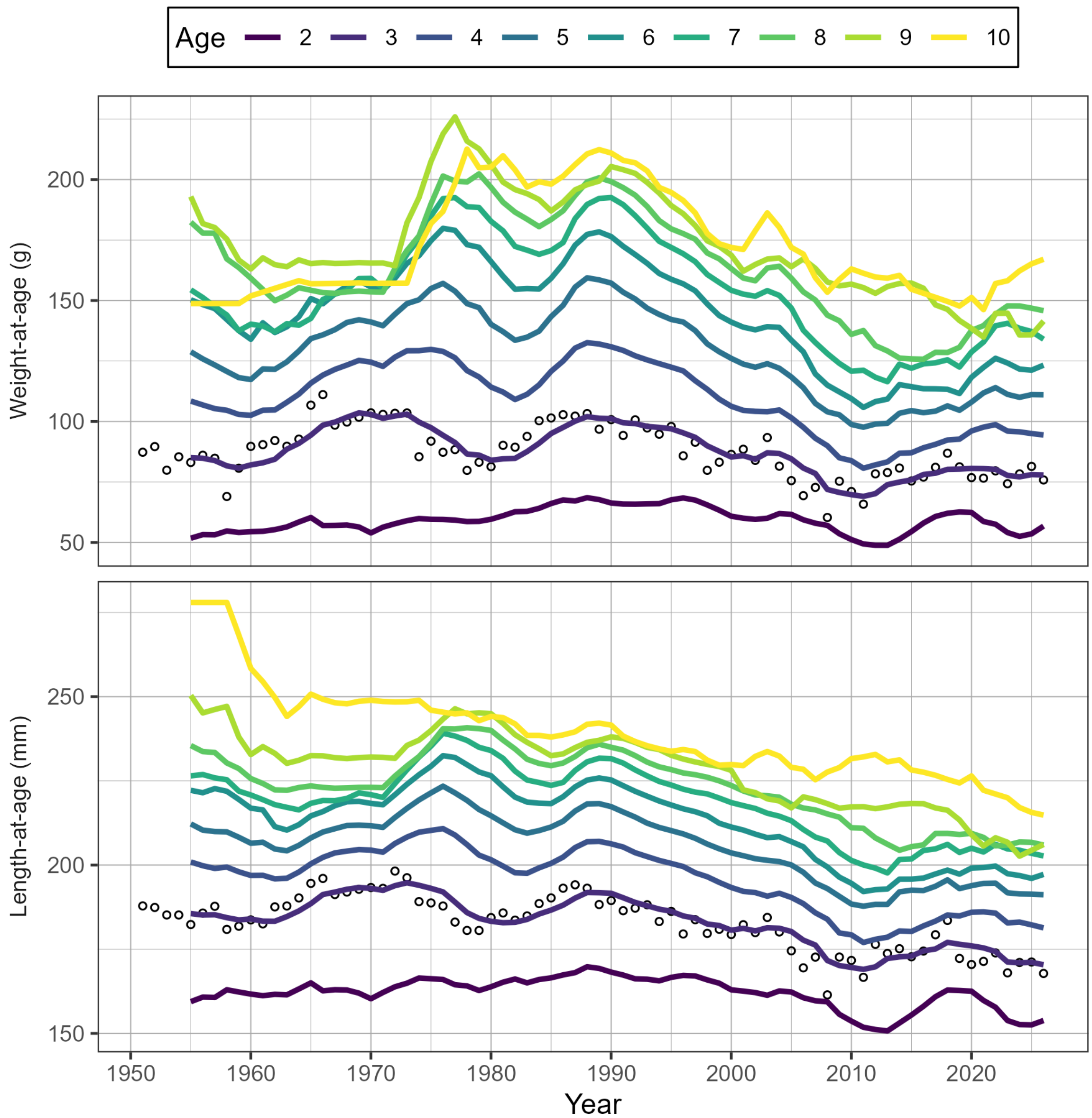

Figure 7. Time series of weight-at-age in grams (g) and length-at-age in milimetres (mm) for age-3 (circles) and 5-year running mean weight- and length-at-age (lines) for Pacific Herring from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). Missing weight- and length-at-age values (i.e., years with no biological samples) are imputed using one of two methods: missing values at the beginning of the time series are imputed by extending the first non-missing value backwards; other missing values are imputed as the mean of the previous 5 years. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a 'plus group' which includes fish ages 10 and older.

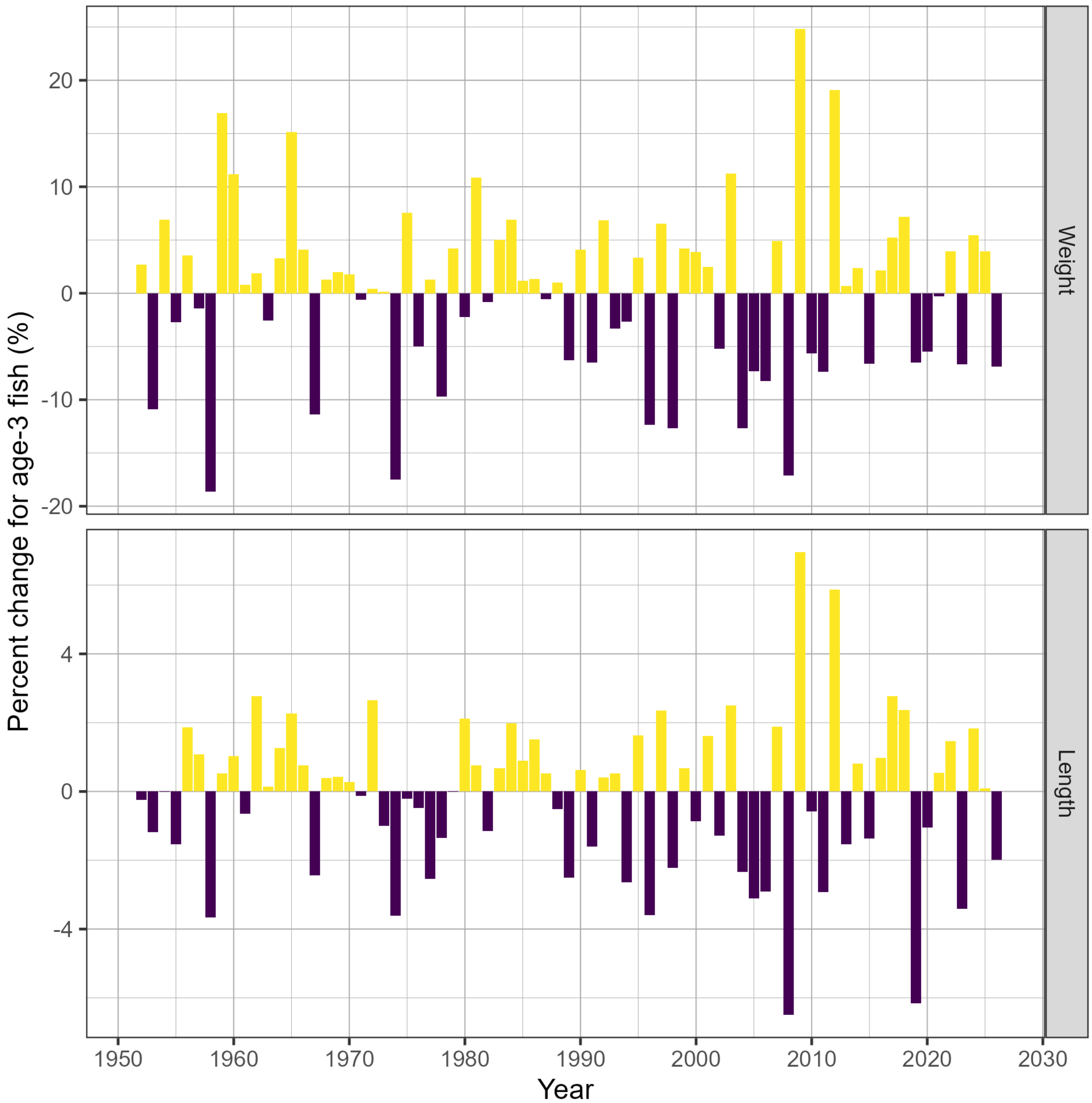

Figure 8. Time series of percent change (%) in weight and length for age-3 fish for Pacific Herring from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-3 fish, respectively, in year t. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

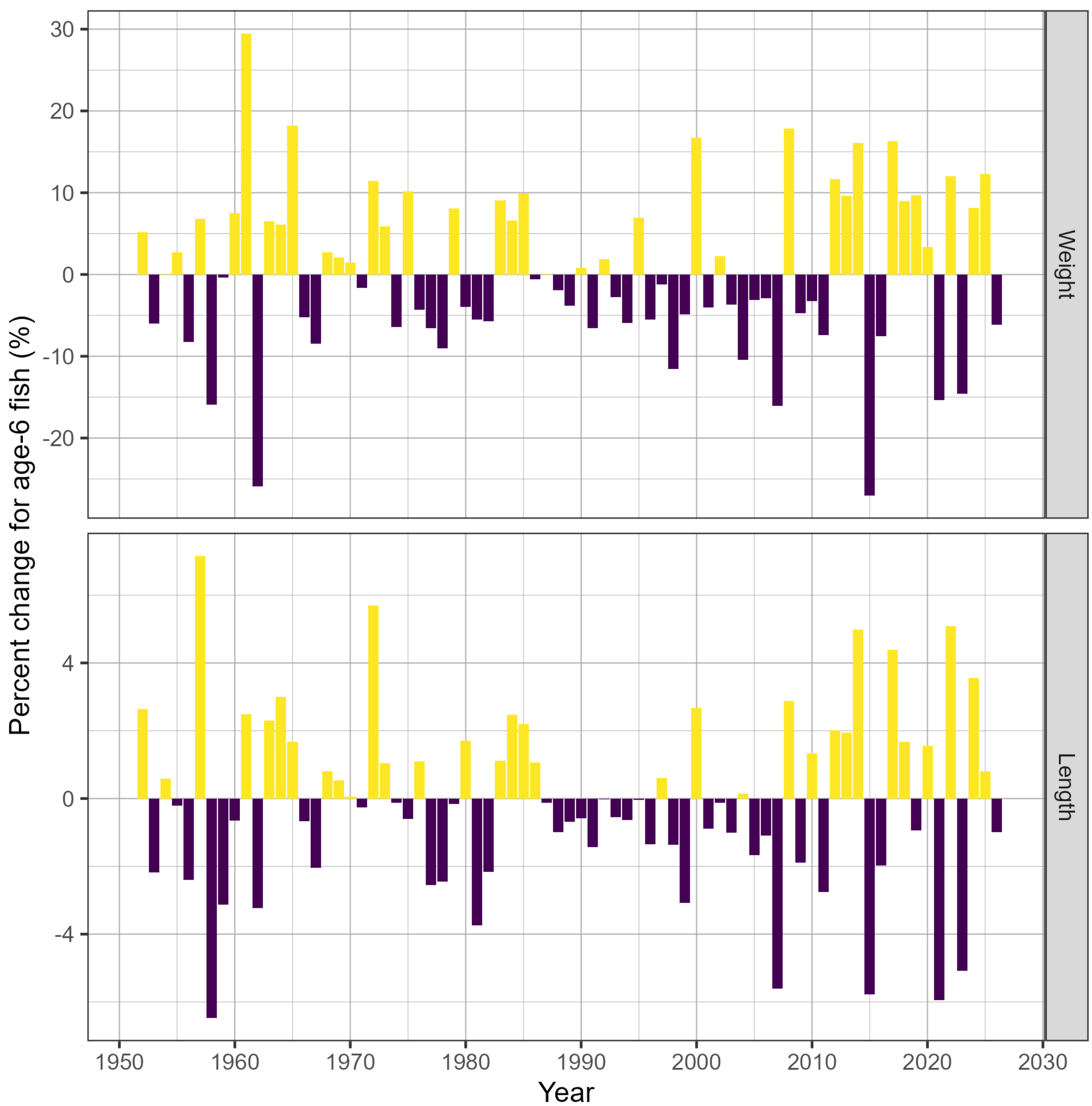

Figure 9. Time series of percent change (%) in weight and length for age-6 fish for Pacific Herring from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-6 fish, respectively, in year t. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

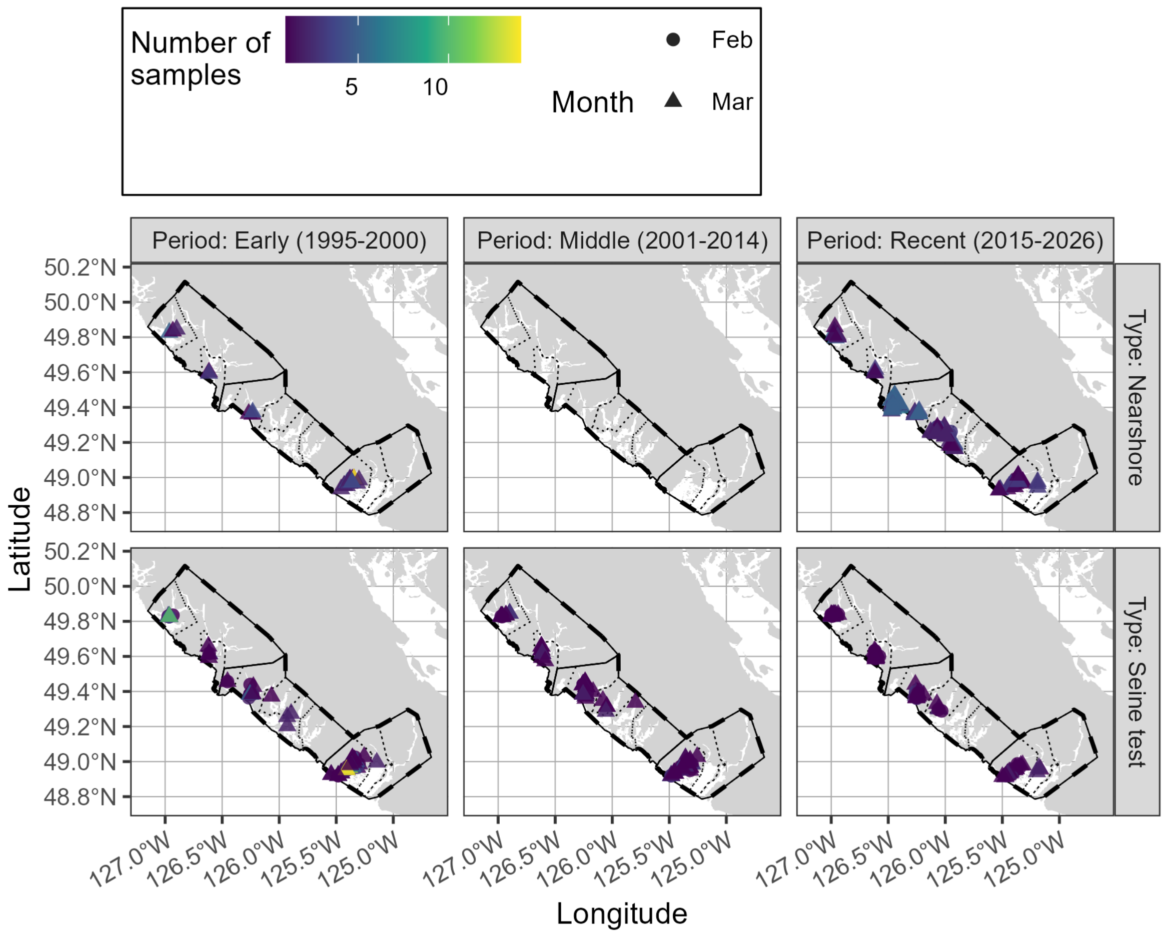

Figure 10. Location of Pacific Herring biological samples in the West Coast of Vancouver Island major stock assessment region (SAR) by period, sample type, and month. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

Figure 11. Number of Pacific Herring samples in the West Coast of Vancouver Island major stock assessment region (SAR) by sample type, statistical area, and gear type. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

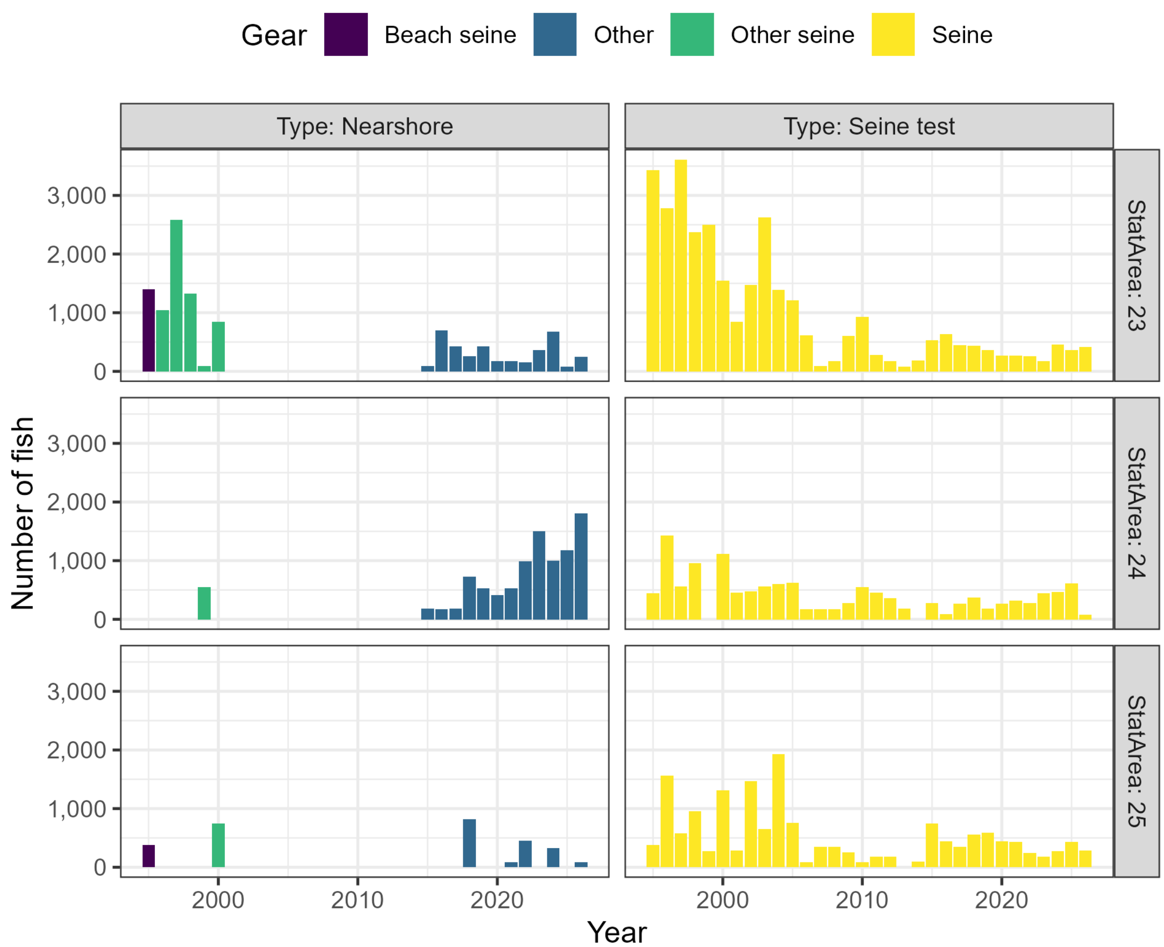

Figure 12. Number of Pacific Herring in biological samples in the West Coast of Vancouver Island major stock assessment region (SAR) by sample type, statistical area, and gear type. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

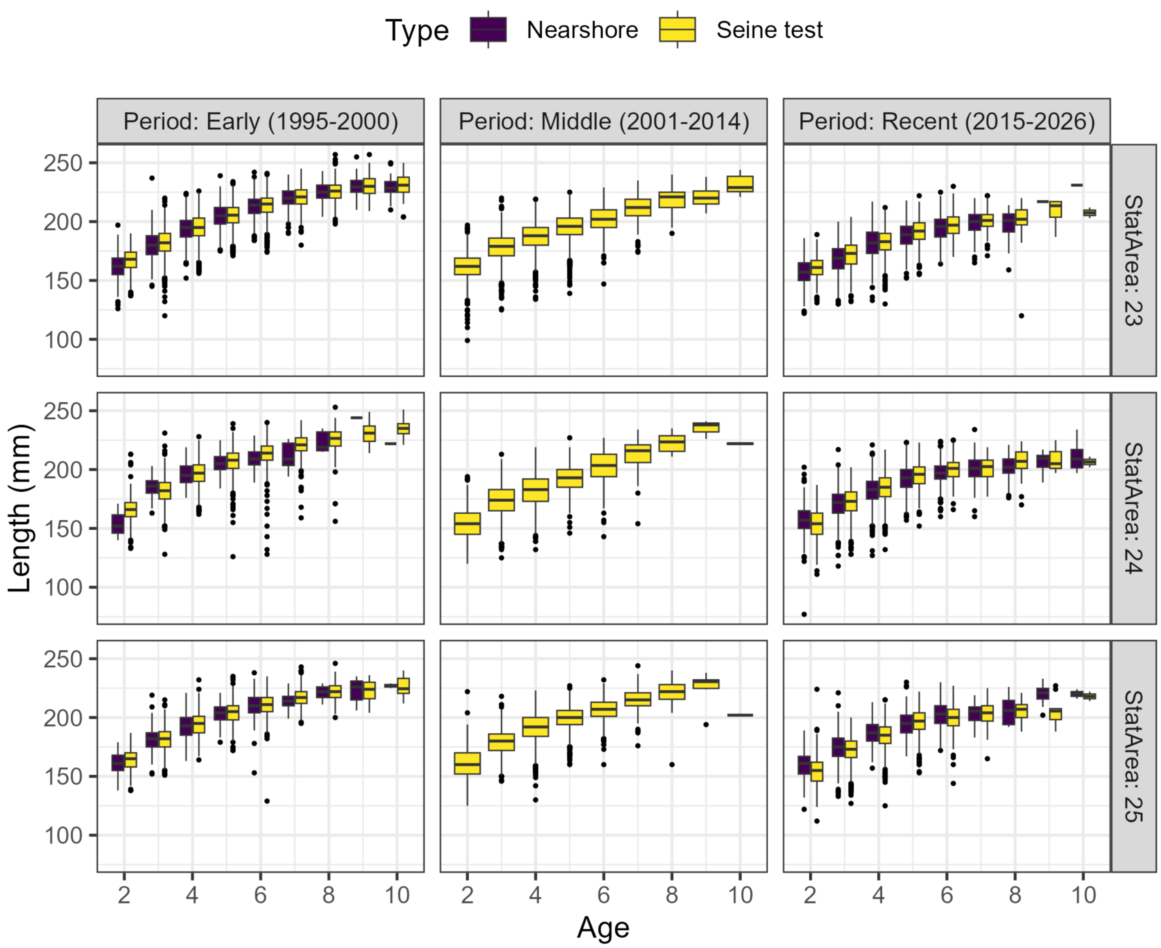

Figure 13. Length-at-age in millimetres (mm) of Pacific Herring in the West Coast of Vancouver Island major stock assessment region (SAR) by period, statistical area, and sample type. The outer edges of the boxes indicate the 25th and 75th percentiles, and the middle lines indicate the 50th percentiles (i.e., medians). The whiskers extend to $1.5 \times$ IQR, where IQR is the distance between the 25th and 75th percentiles, and dots indicate outliers. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations. The age-10 class is a 'plus group' which includes fish ages 10 and older.

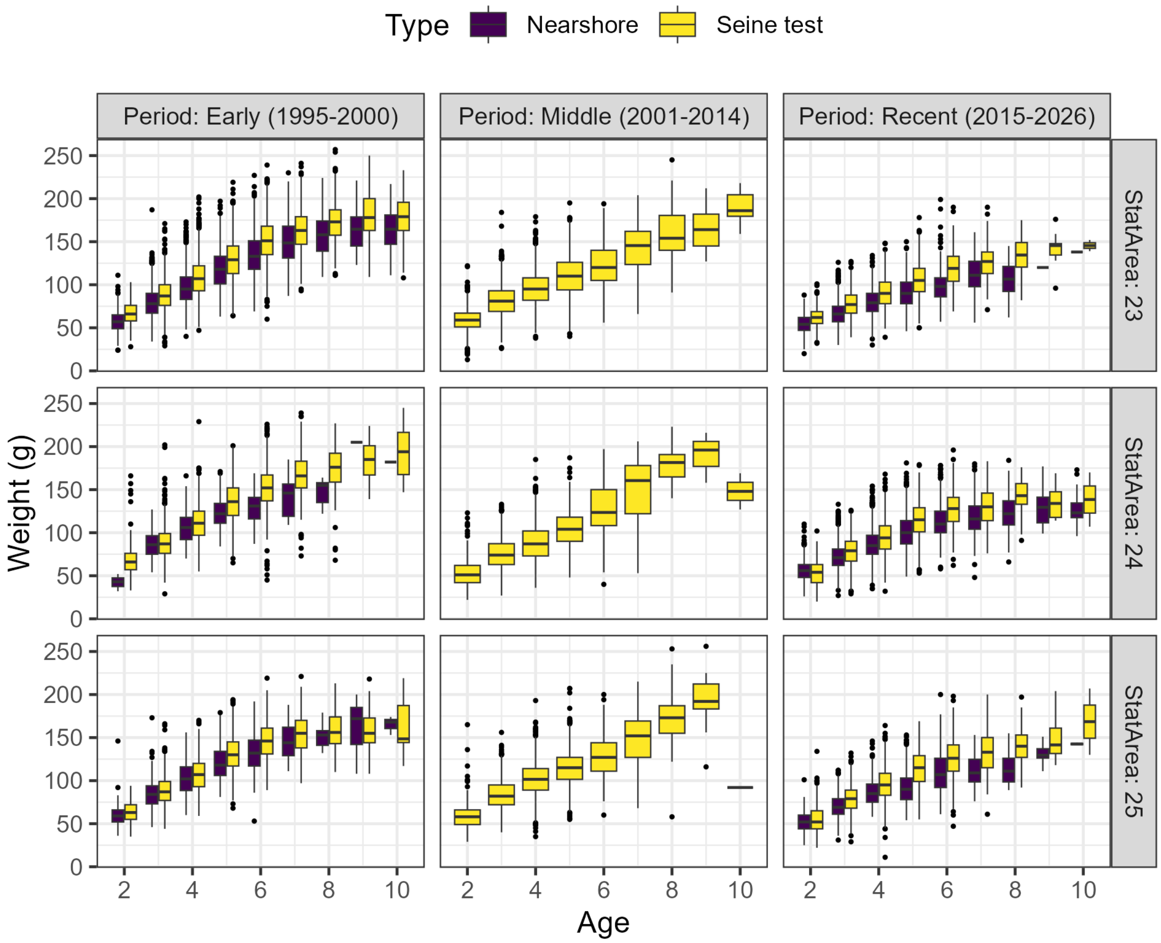

Figure 14. Weight-at-age in grams (g) of Pacific Herring in the West Coast of Vancouver Island major stock assessment region (SAR) by period, statistical area, and sample type. The outer edges of the boxes indicate the 25th and 75th percentiles, and the middle lines indicate the 50th percentiles (i.e., medians). The whiskers extend to 1.5 $\times$ IQR, where IQR is the distance between the 25th and 75th percentiles, and dots indicate outliers. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations. The age-10 class is a 'plus group' which includes fish ages 10 and older.

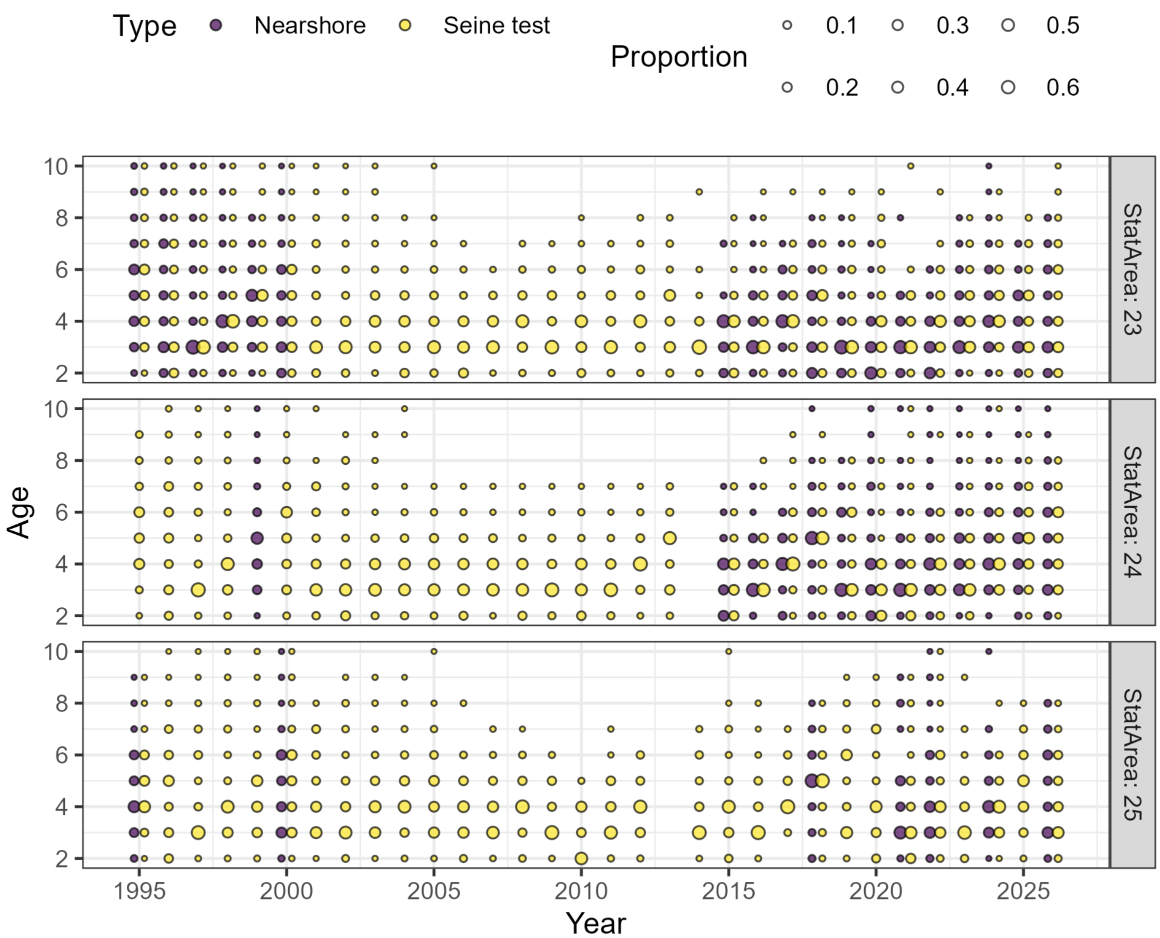

Figure 15. Proportion-at-age of Pacific Herring in the West Coast of Vancouver Island major stock assessment region (SAR) by statistical area and sample type. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

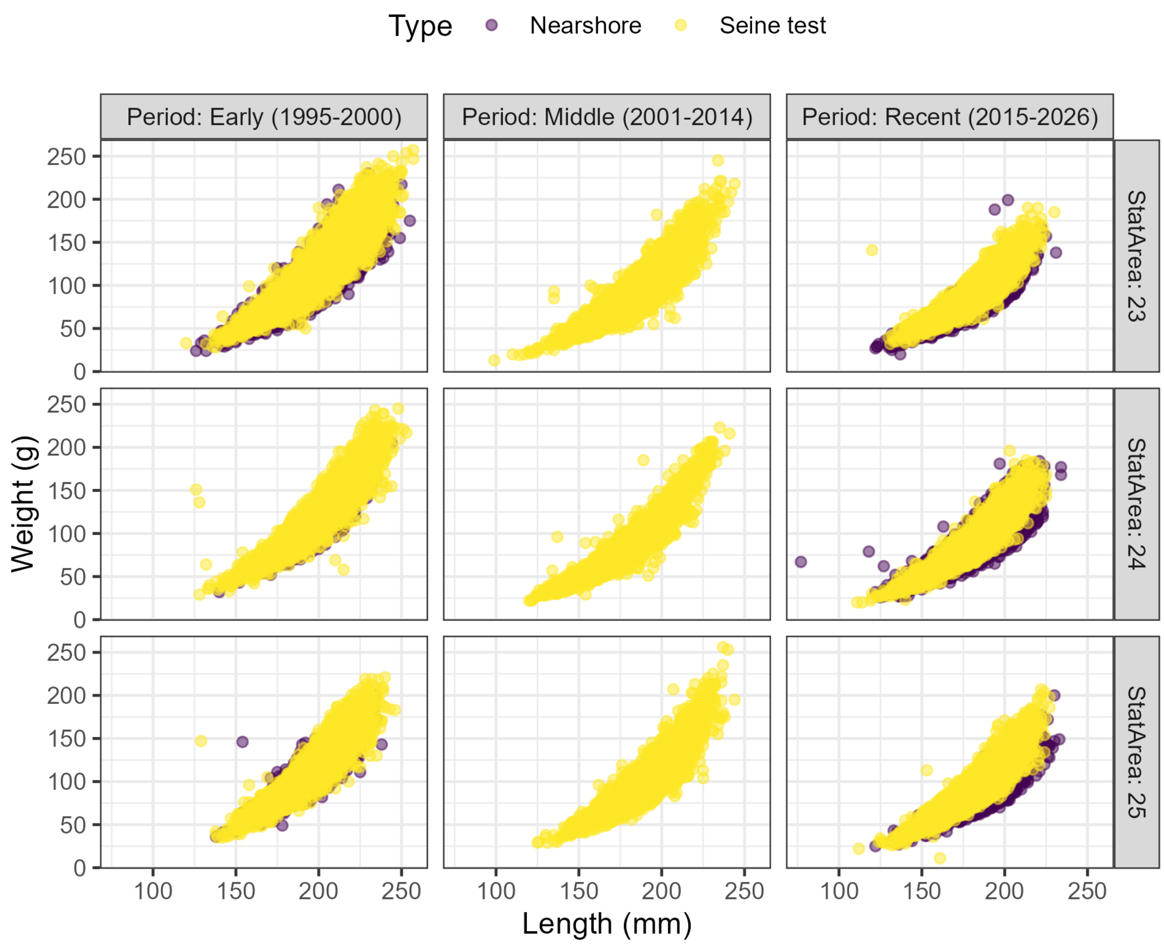

Figure 16. Length-weight relationship for Pacific Herring in the West Coast of Vancouver Island major stock assessment region (SAR) by period, statistical area, and sample type. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

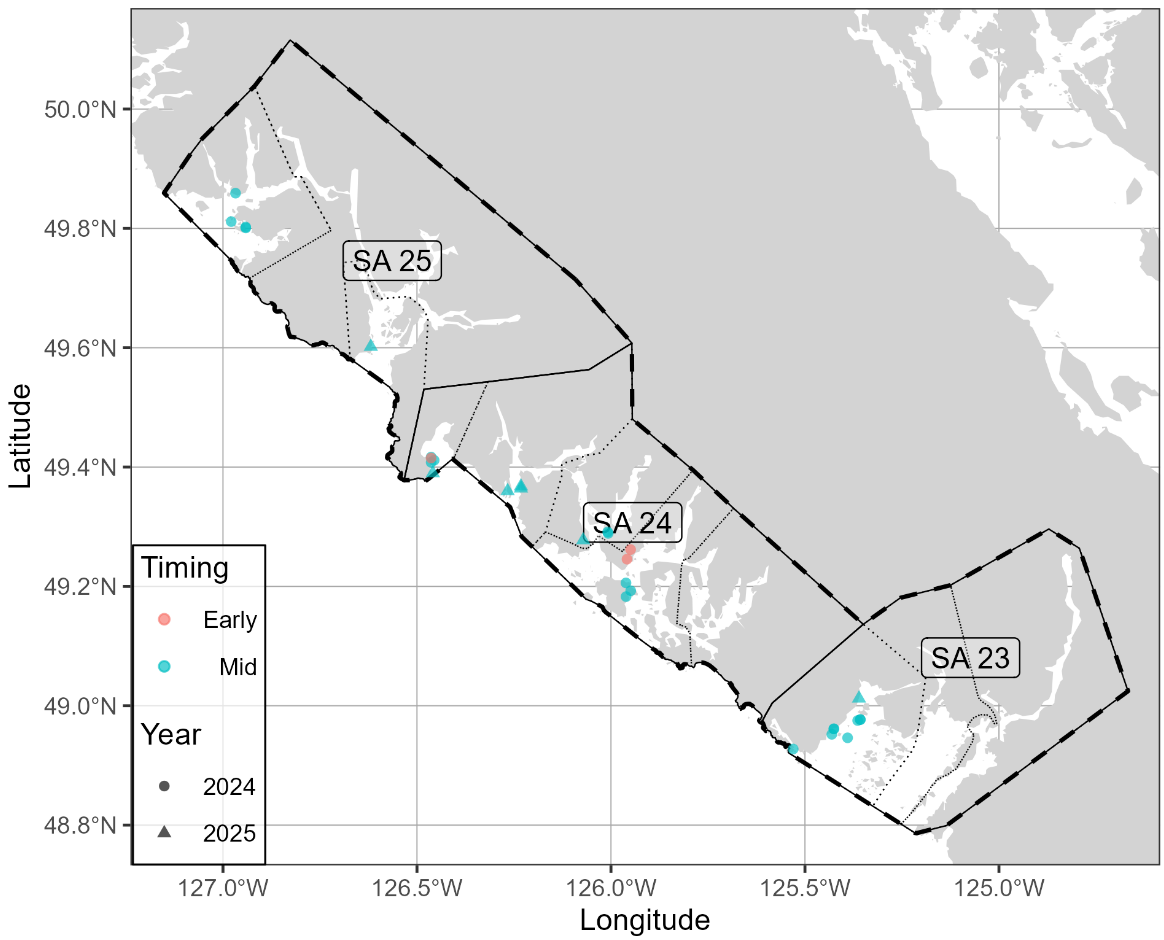

Figure 17. Nearshore genetics sampling in the West Coast of Vancouver Island major SAR. Note that timing indicates the collection date: 'Early' is January and February, 'Mid' is March and April, and 'Late' is May, June, and July. In addition, note that genetic samples from 2025 are not yet available. See Figure 2 for legend.

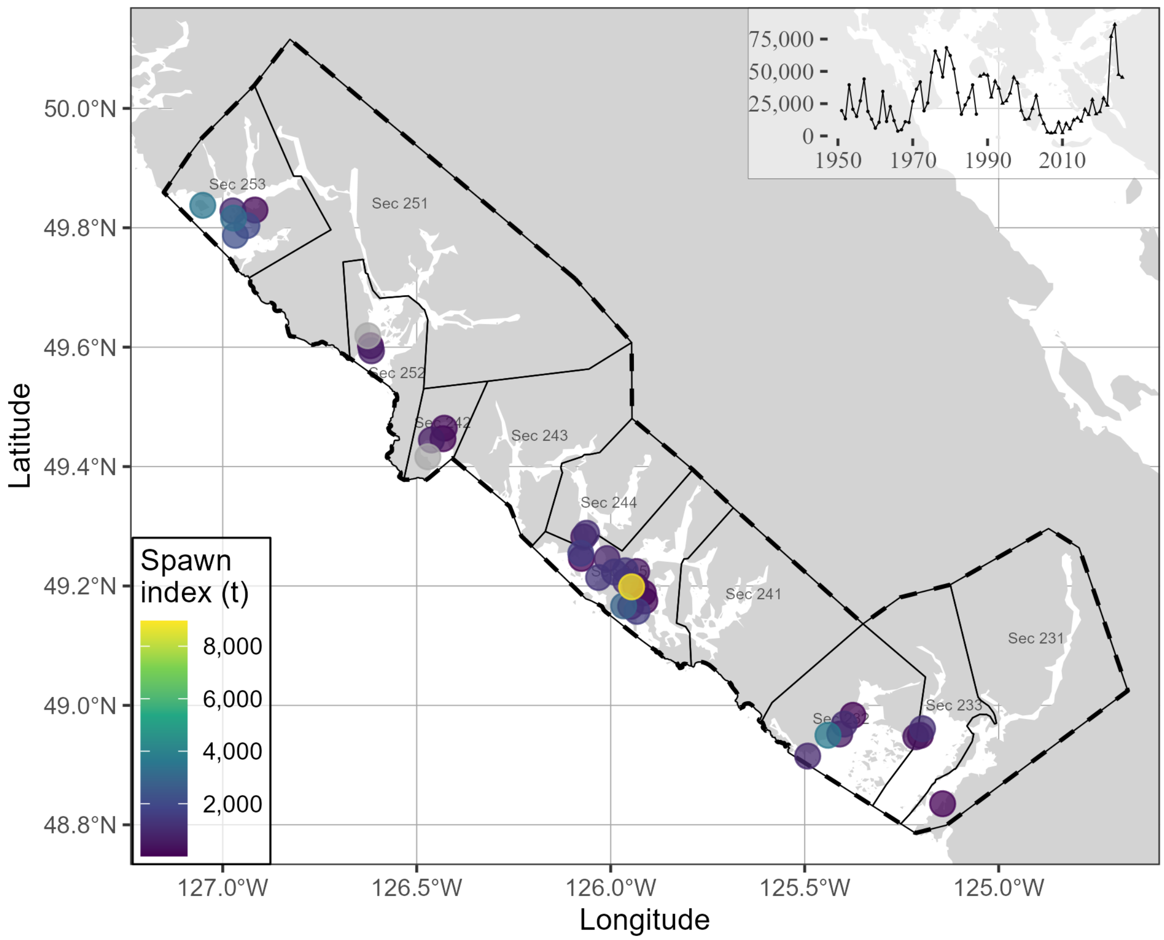

Figure 18. Pacific Herring spawn survey locations, and spawn index in metric tonnes (t) in 2026 in the West Coast of Vancouver Island major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km).

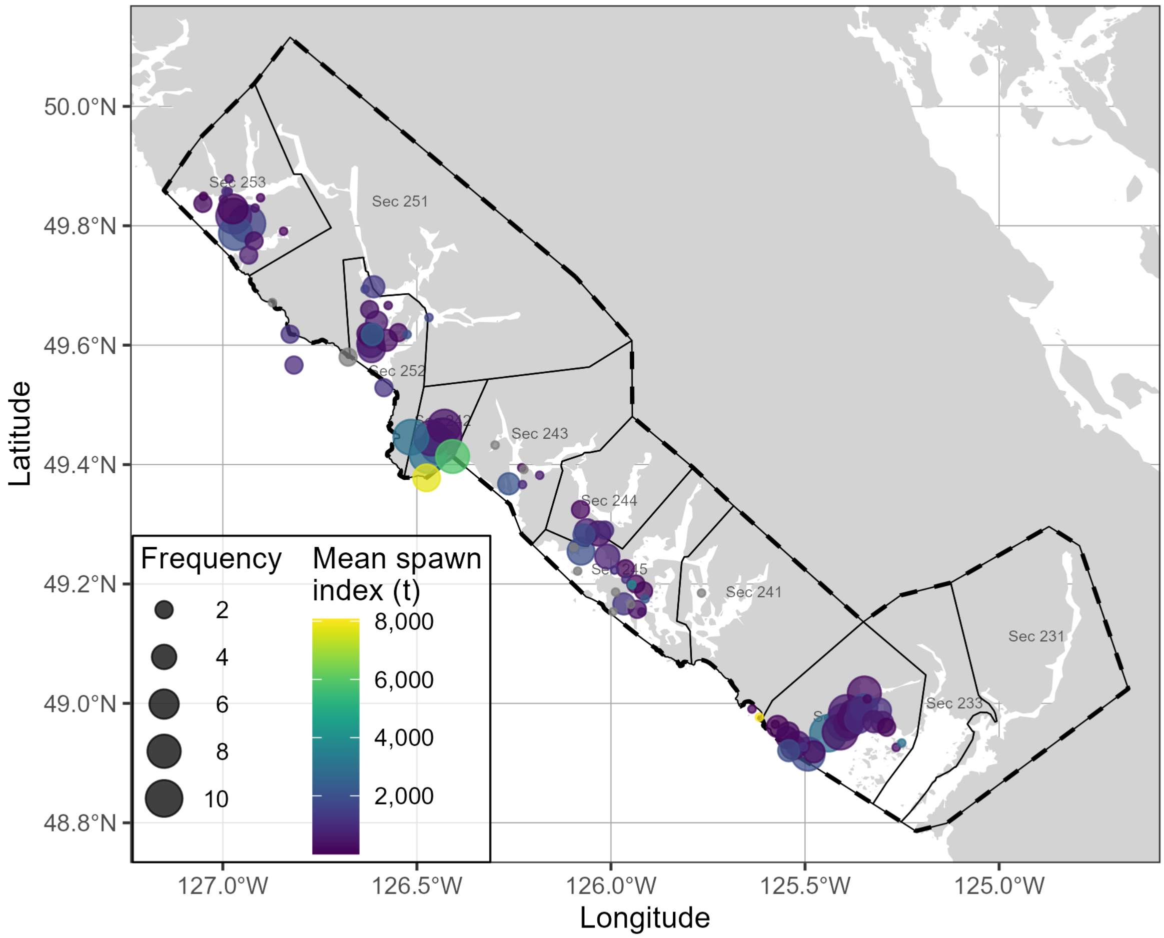

Figure 19. Pacific Herring spawn survey locations, mean spawn index in metric tonnes (t), and spawn frequency from 2016 to 2025 in the West Coast of Vancouver Island major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Missing spawn index values indicate incomplete surveys (grey circles). Units: kilometres (km).

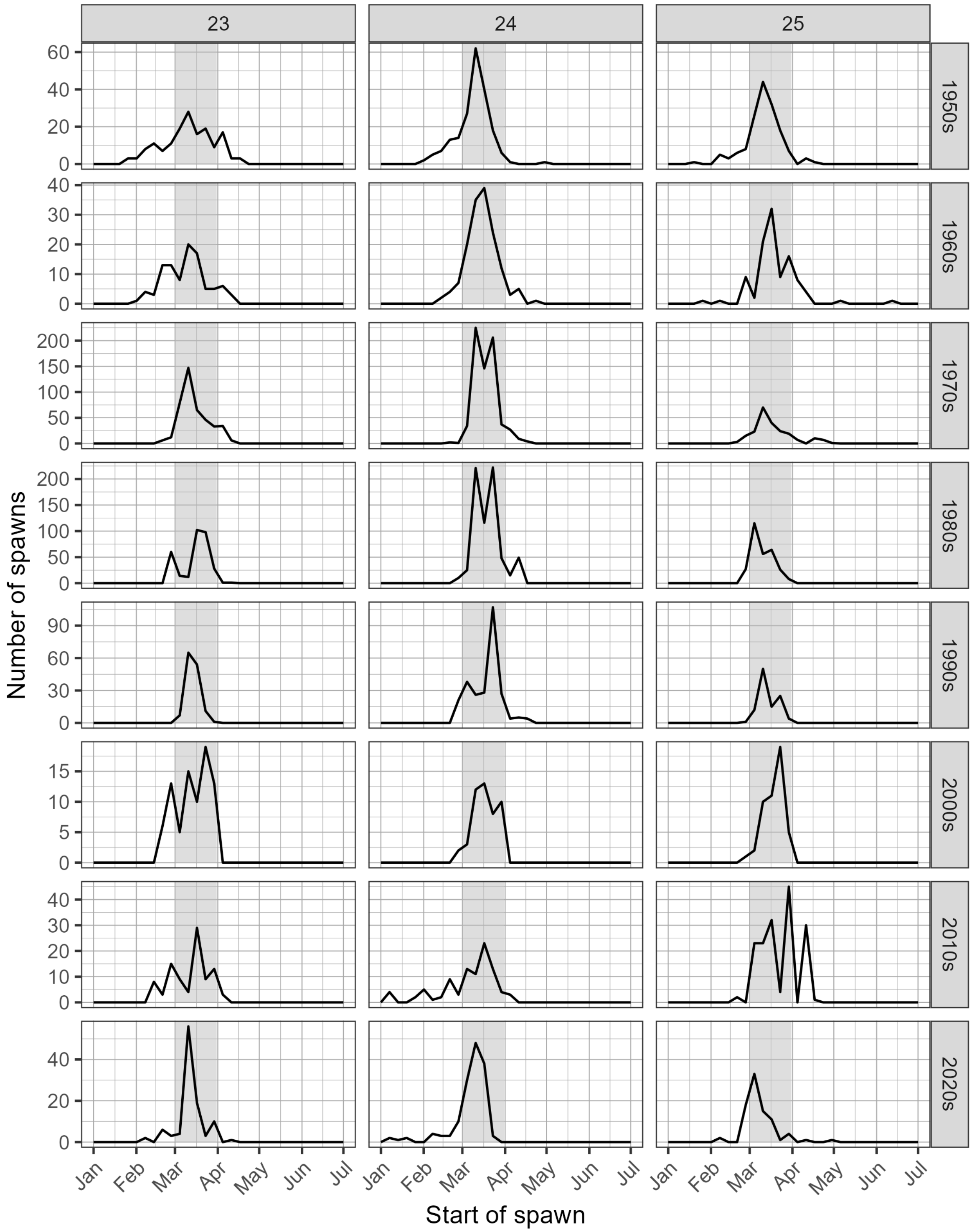

Figure 20. Pacific Herring spawn start date by decade and Statistical Area. Grey shaded regions indicate March 1st to 31st. Note that spawn size and intensity varies; therefore the number of spawns is not directly proportional to spawn extent or biomass.

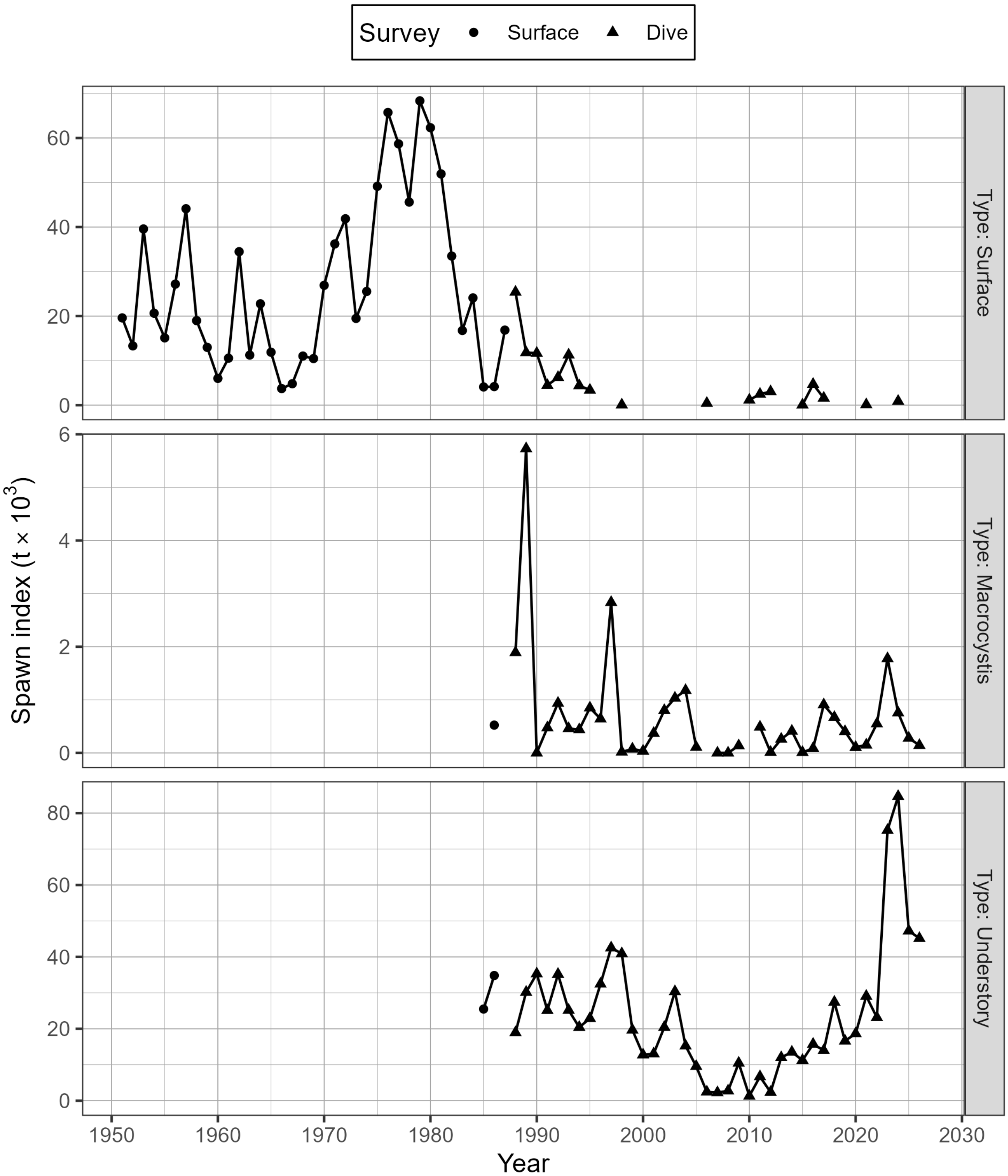

Figure 21. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) by type for Pacific Herring from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). There are three types of spawn survey observations: observations of spawn taken from the surface usually at low tide, underwater observations of spawn on giant kelp, Macrocystis (*Macrocystis* spp.), and underwater observations of spawn on other types of algae and the substrate, which we refer to as 'understory.' The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). See Figure 22 for the spawn survey method.

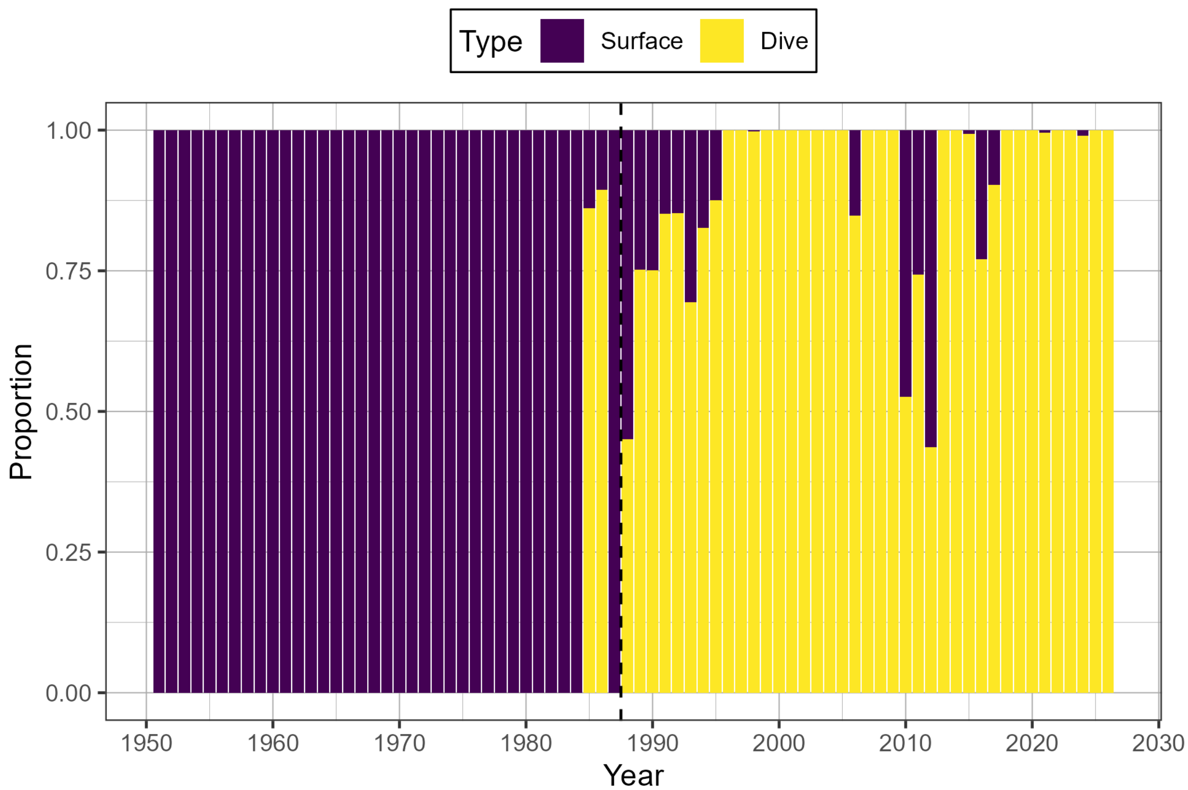

Figure 22. Time series of proportion of spawn index by surface and dive surveys for Pacific Herring from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

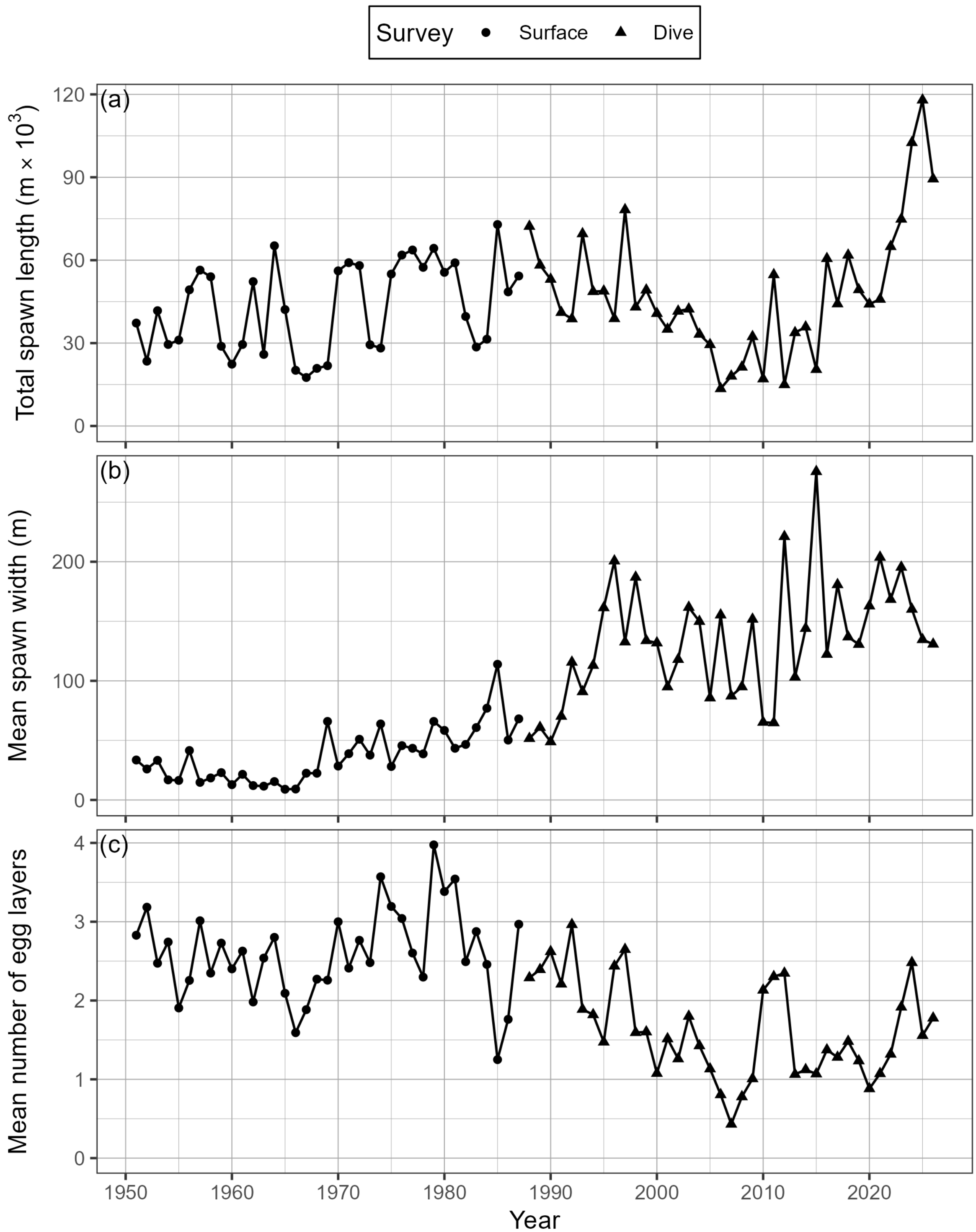

Figure 23. Time series of total spawn length in thousands of metres (m $\times$ 10^3; panel a), mean spawn width in metres (b), and mean number of egg layers (c) for Pacific Herring from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

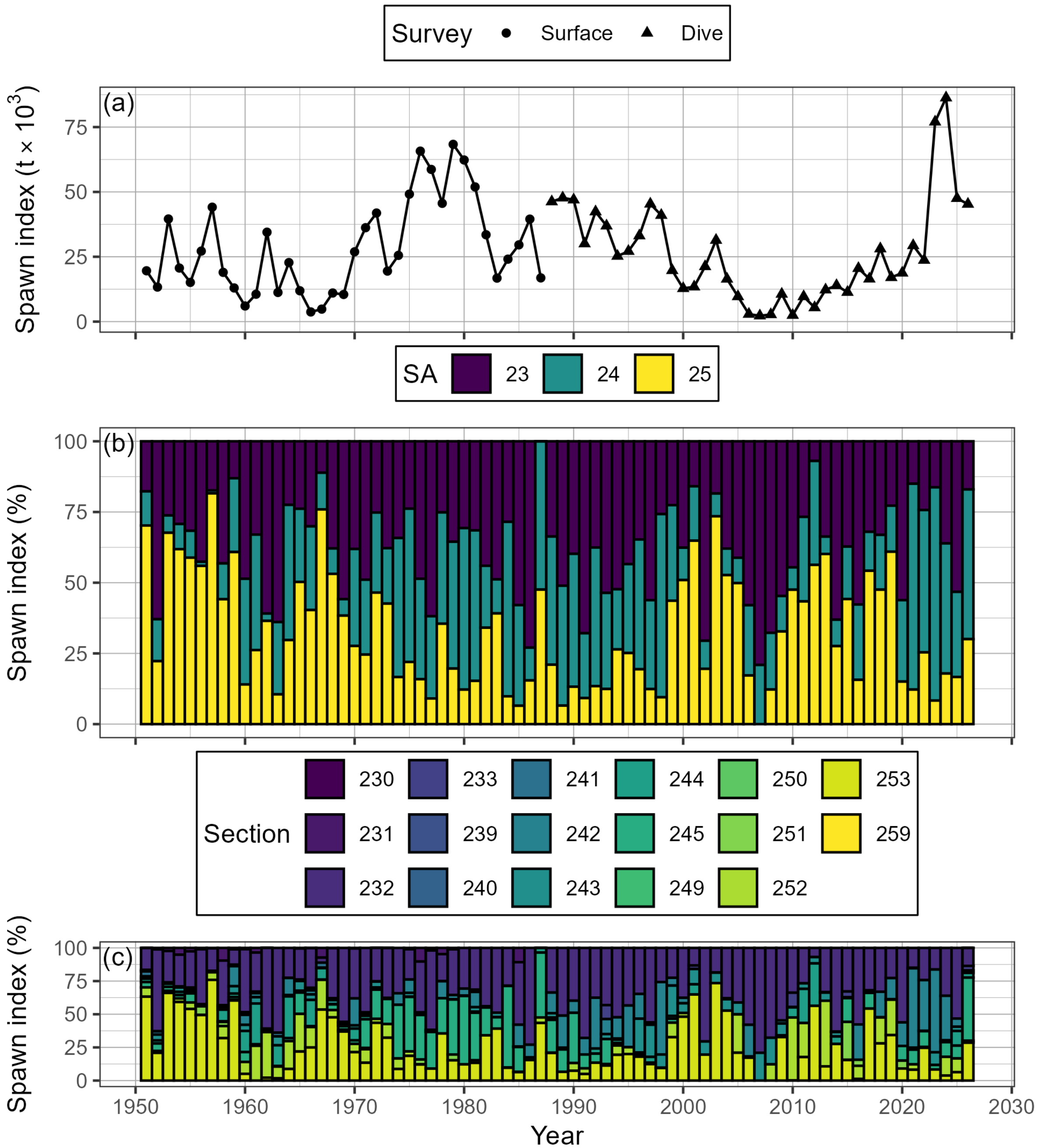

Figure 24. Time series of spawn index in thousands of metric tonnes (t $\times$ 10^3) for Pacific Herring from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR; panel a), as well as percent contributed by Statistical Area (SA), and Section (b, & c, respectively). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). Note that spawn surveys in the dive survey period (1988 to 2026) are a combination of surface and dive surveys (Figures 21 and 22). The 'spawn index' is not scaled by the spawn survey scaling parameter, q.

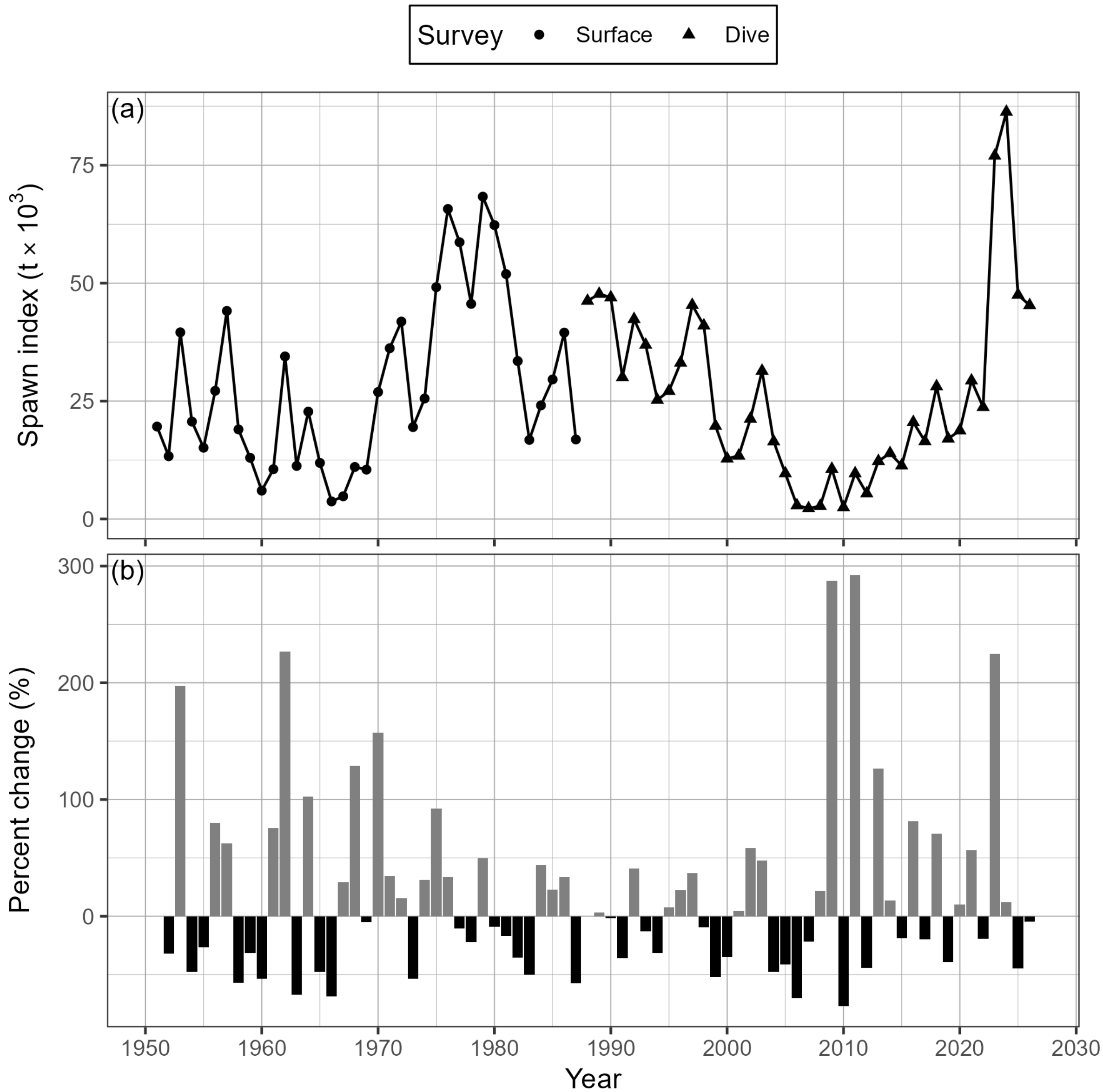

Figure 25. Time series of spawn index in thousands of metric tonnes (t $\times$ 10^3) for Pacific Herring from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR; panel a), and percent change (b). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the spawn index in year t. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Note that spawn surveys in the dive survey period (1988 to 2026) are a combination of surface and dive surveys (Figures 21 and 22).

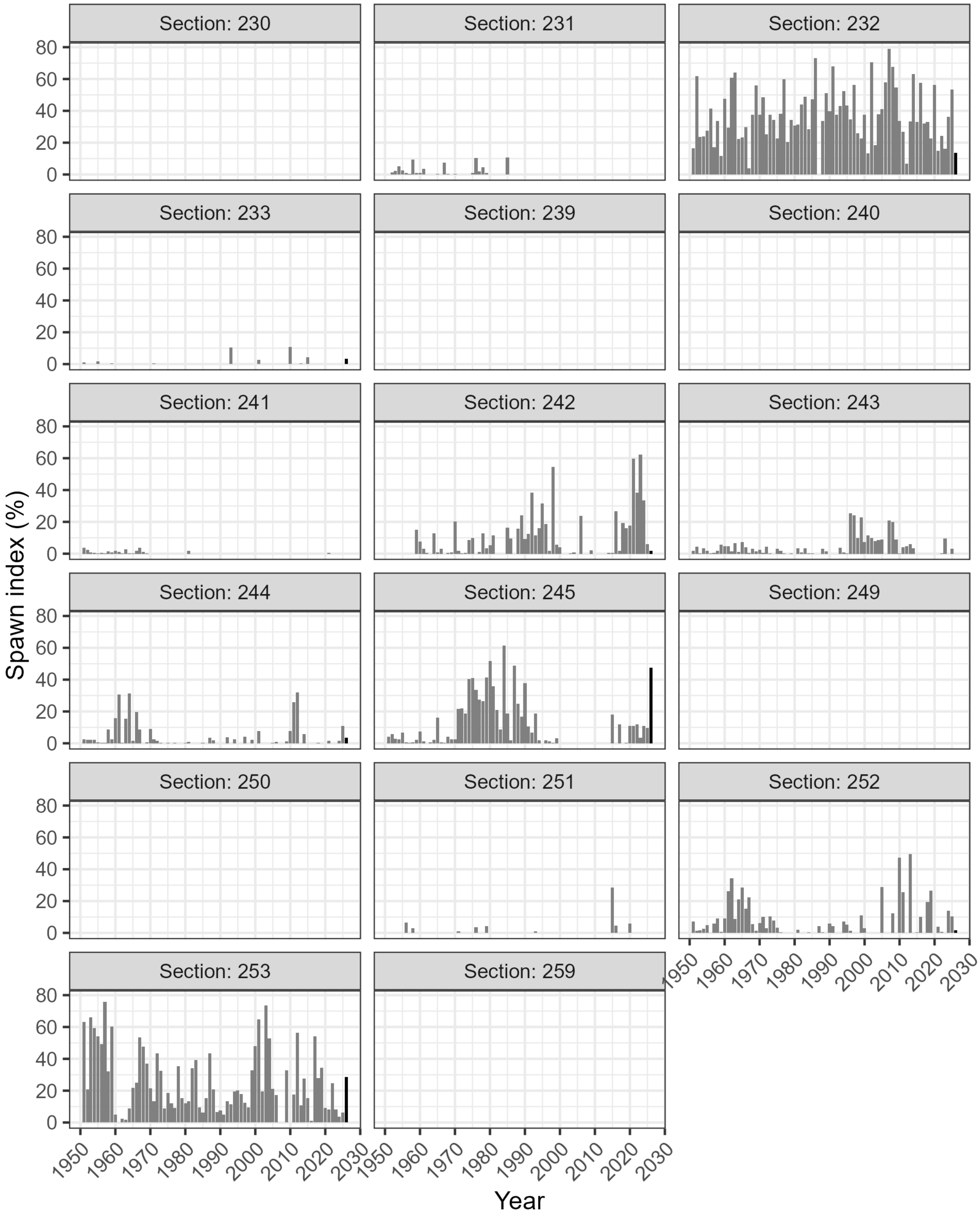

Figure 26. Time series of percent of spawn index by Section for Pacific Herring from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The year 2026 has a darker bar to facilitate interpretation. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The 'spawn index' is not scaled by the spawn survey scaling parameter, q.

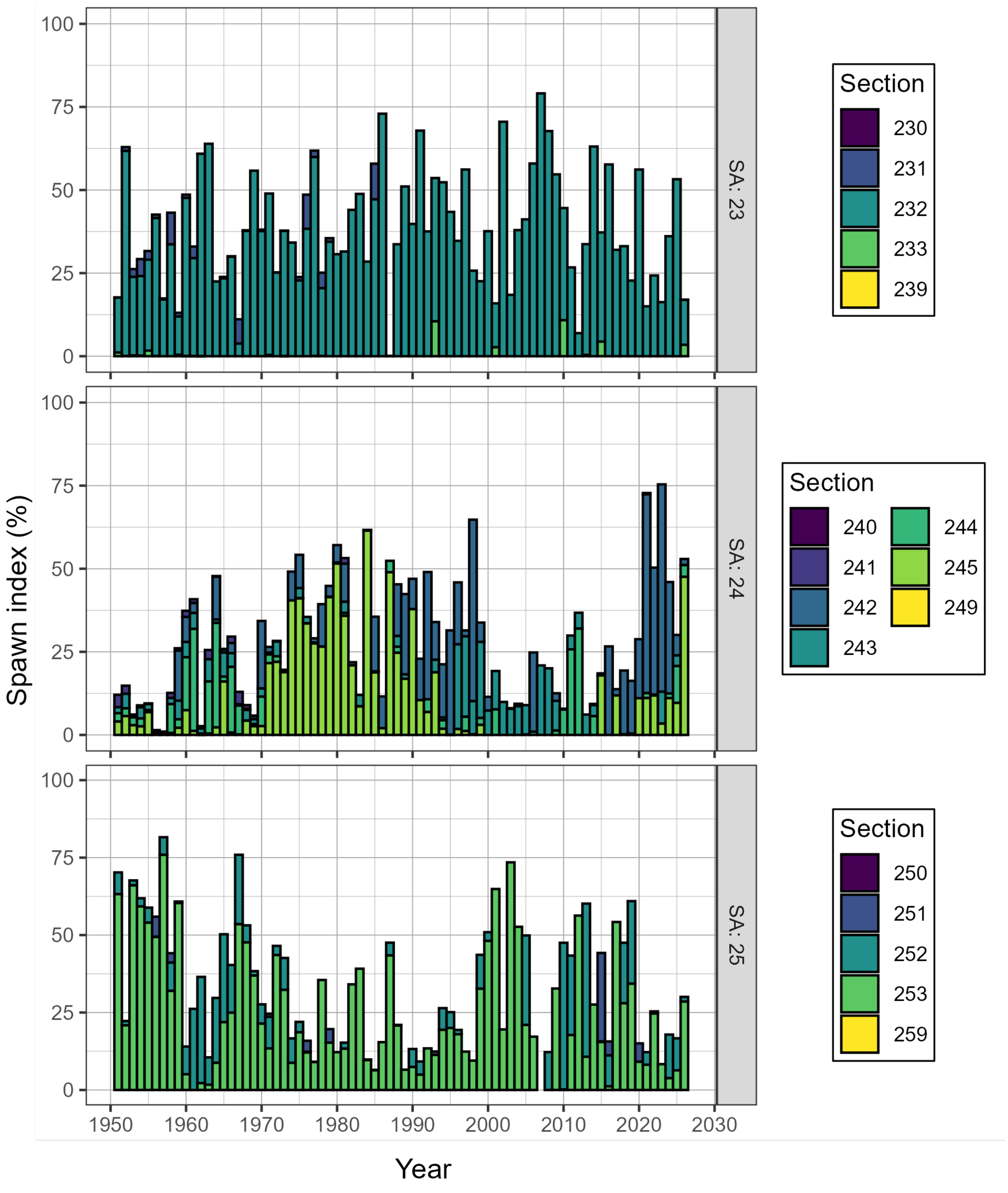

Figure 27. Time series of percent of spawn index by Statistical Area (SA) and Section for Pacific Herring from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The 'spawn index' is not scaled by the spawn survey scaling parameter, q.

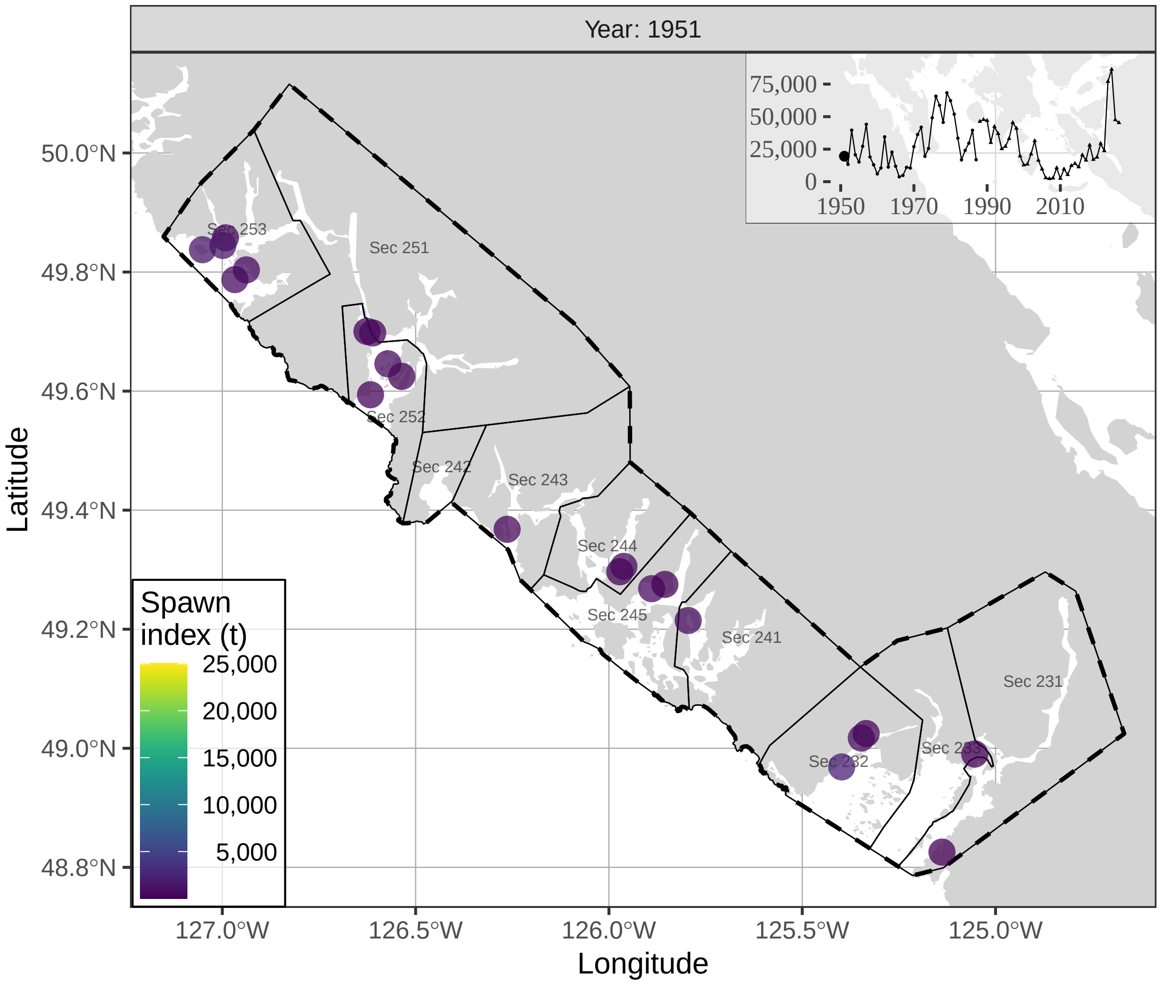

Figure 28. Animation of Pacific Herring spawn index in metric tonnes (t) by Location from 1951 to 2026 in the West Coast of Vancouver Island major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km). View the animation: download the report, open with Adobe, enable Java, and click "play".